

Gen Z Adoption of AI in Banking: Evidence from Malaysia
An Extended Technology Acceptance Model Using PLS-SEM
(Original thesis: Banking 4.0 in Malaysia: Is Gen Z Ready for It?)

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ABSTRACT

This study investigates the factors that influence Gen Z Malaysians' intention to adopt AI in banking. A total of 633 responses from Gen Z Malaysians were collected through a questionnaire. The extended Technology Acceptance Model (TAM) was employed. Variables incorporated are perceived usefulness, perceived ease of use, attitude towards AI in banking, perceived risk, and perceived trust. Reliability and validity tests confirmed the constructs' reliability and convergent and discriminant validity, as well as the absence of common-method bias. PLS-SEM is used to analyse path relationships and test hypotheses. The results indicate that all hypothesised relationships are significant. Perceived usefulness has a direct and indirect positive influence on the intention to adopt AI in banking, mediated by attitude. Perceived ease of use positively and indirectly influences intention through attitude and perceived usefulness. Perceived risk negatively influences intention through attitude. Perceived trust positively influences intention by shaping attitude. By examining the effect sizes of each variable, the attitude that serves as the mediator across all variables has the greatest impact on intention. Following that, the ranking by effect size is perceived usefulness, perceived ease of use, perceived trust, and finally, perceived risk. The findings suggested that promoting trust through risk reduction and enhancing the benefits and convenience of banking AI are critical to stimulating adoption.

Keywords: Banking 4.0, Artificial Intelligence (AI), Technology Acceptance Model (TAM), Gen Z, FinTech

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LIST OF ABBREVIATIONS

TAM	Technology Acceptance Model
TRA	Theory of Reasoned Action
TPB	Theory of Planned Behaviour
PU	Perceived usefulness
PEOU	Perceived ease of use
ATT	Attitude towards a particular technology
PR	Perceived risk
PT	Perceived trust
INT	Intention to adopt AI in banking
EFA	Exploratory Factor Analysis
CFA	Confirmatory Factor Analysis
SEM	Structural Equation Modelling
CB-SEM	Covariance-based SEM
PLS-SEM	Partial Least Squares SEM

CHAPTER 1

INTRODUCTION

1.0 Introduction

Chapter 1 outlines the study's framework. Section 1.1 offers the background for the study, while Section 1.2 articulates the problem statement. In Section 1.3, the study's significance is discussed. The research questions and objectives are revealed in Sections 1.4 and 1.5. Finally, Section 1.6 illustrates the overall organisation for the subsequent content.

1.1 Background of the Study

The global production landscape has undergone a substantial metamorphosis as it enters a new era of Industry 4.0. This revolution is driven by several technological advancements, including enhancements in data and connectivity, as well as the integration of machines and robots into the production process. (McKinsey & Company, 2022). As the pioneer in the market, the financial industry is the trailblazer in adopting 4IR technology. The banking industry is no exception. It has been undergoing a disruptive revolution, known as “Banking 4.0 – Banking Everywhere but Never at a Bank,” since 2017, following the conclusion of the Banking 3.0 period. For instance, mobile wallets are becoming increasingly common, fintech usage worldwide is normalised, and Artificial Intelligence (AI) is integrated into the banking model. The Banking 4.0 revolution is an ongoing process that is drastically reshaping the financial industry and is anticipated to persist until 2050 (PwC, n.d.).

The banking industry has undergone three revolutions, beginning with Banking 1.0, which occurred four decades ago and was marked by the adoption of mainframe systems for transaction recording. Banking 2.0 emerged in the 1990s, driven by the creation of compatible open systems that encouraged banks worldwide to adopt fourth-generation language (4GL) technologies in their recording systems. Banking 3.0 was initiated by the emergence of the Internet and mobile technology, which facilitates large-scale electronic payments. It is defined as the banking business model that

provides customers with a streamlined, cross-channel self-service payments experience (Bhatia, 2022).

Although Banking 3.0 helps banks manage high-volume, straightforward processes, it still faces certain primary shortcomings. Firstly, third-generation banks focused on self-service options rather than providing a digital experience. The channels they created lack flexibility and configurability. Secondly, these banks underinvested in their back- and mid-office, where manual interventions, reconciliation, and scalable maintenance are still required in their operational processes. Thirdly, data and application duplication are significant problems among third-generation banks, primarily due to the use of monolithic applications and technological constraints. This has resulted in over 90% data definition duplication, over 80% data duplication, and over 50% application redundancy. Third-generation banks often lack a compelling business case for investing in technology to address these shortcomings. Their existing project-based funding structures deter them from undergoing a generational shift towards the fourth generation, as such a shift requires scalable, immediate investment but does not yield immediate benefits. These banks also operate in a pressured and constrained business environment, making paying for this ‘Technical Debt’ difficult. All these shortcomings have fuelled the emergence of the Banking 4.0 revolution, which seeks to address them (Bhatia, 2022).

Banking 4.0’s central strategies involve utilising 4IR technologies for asset digitalisation, digital identity creation, the provision of customer-exclusive offers, and service customisation. Mehdiabadi et al. (2020) state that it provides customers with a personalised, integrated, and customised experience that alters the interactions between customers and banks. The vision of Banking 4.0 is to industrialise the banking process, promoting scalable, flexible and robust business results.¹ In addition, Banking 4.0 requires continuous investment in building a comprehensive banking AI corpus and algorithms to transform traditional banking, as well as in large-scale

¹ Some initiatives include industrialising the data platform to leverage cloud technologies, the Financial Crimes Management Process 3.0, Risk Management 3.0, and building Application Programming Interfaces (APIs) to integrate applications and data.

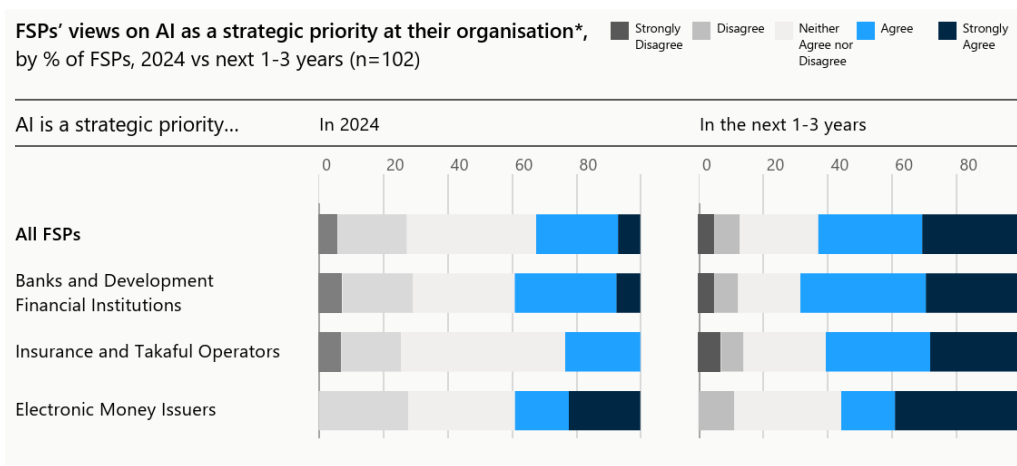
customisation and the systematisation of document management (Noreen et al., 2023). Following Bhatia (2022), Banking AI is also one of the six forces of Banking 4.0 industrialisation and innovation.²

Belanche et al. (2019) argue that AI has become a megatrend reforming all sectors. The banking industry also considers AI implementation as part of its business strategy to remain competitive in a dynamic environment. Many banks aim to replace costly, monotonous, and repetitive activities with AI (Kaya et al., 2019). Vijai (2019) proposed several basic AI applications in the banking industry, including chatbots, personalised services, and AI-aided self-service, which can create value beyond conventional human advisory services. The benefits of AI applications in banks to prevent fraud (Da Silva, 2021; Narang et al., 2024; Roseline et al., 2022), increase accuracy and reliability (Maja & Letaba, 2022), provide swift services (Garg et al., 2020), and secure banking services have made it an inevitable transformation in the industry.

Bank Negara Malaysia (2025) has conducted the BNM AI Survey 2024, targeting 120 financial service providers (FSPs) in Malaysia. The result shows that 71% of banking institutions and development financial institutions have implemented AI applications. This is a 56% increase compared to last year. Additionally, 77% of insurance and takaful operators in Malaysia have adopted AI applications, a 58% increase from last year. Notably, over 60% of the FSPs emphasised AI as a strategic priority at their institutions for the next 1 to 3 years (Figure 1.1).

² The other five are technology investments, digitised experience, embedded innovation, intelligent process and technology, and industrialised data.

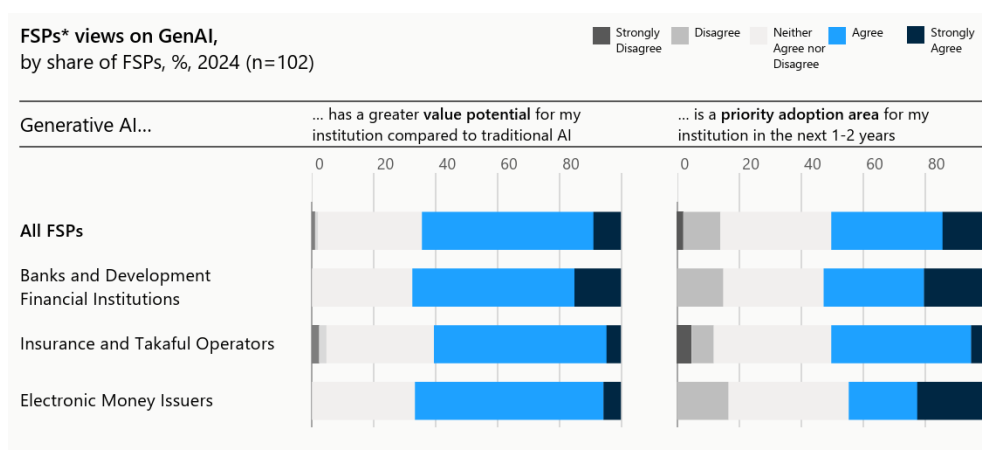
Figure 1.1: FSP's views on AI as a strategic priority at their organisation.



Source: Bank Negara Malaysia (2025)

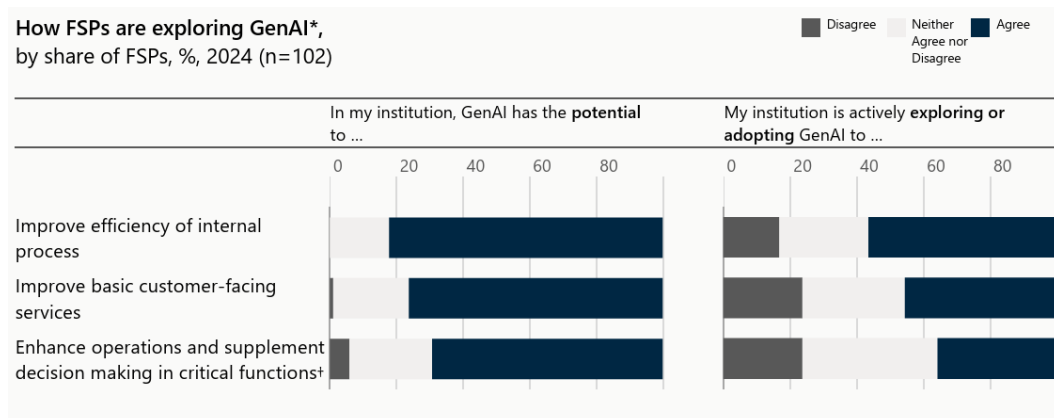
Generative AI (GenAI) has been a primary focus for FSPs due to its potential benefits for their institutions. The survey results indicate that more than 60% of FSPs agree that GenAI has greater potential value than traditional AI. Approximately one-half of the FSPs mentioned GenAI as their priority adoption area for the upcoming two years (Figure 1.2). There are three focuses for realising GenAI's potential within the institution, as outlined in the survey. Over 80% of the FSPs believe that GenAI can improve efficiency and internal processes, and over 75% stated that it improves basic customer-facing services. Nearly 70% agree that it can enhance operations and supplement decision-making in critical functions. However, the disagreement in whether their institution is actively exploring or adopting GenAI is significant, especially for the latter two potential (Figure 1.3).

Figure 1.2: FSPs' view on GenAI.



Source: Bank Negara Malaysia (2025)

Figure 1.3: How FSPs are exploring GenAI.



Source: Bank Negara Malaysia (2025)

In summary, Banking 4.0 requires modularising technical estates and leveraging new technologies continuously. Additionally, initiatives to modernise applications, eliminate outmoded technology, build an integrated layer using APIs, adopt cloud-native technology, and develop AI technology and conversational systems will ultimately shift banks into the Banking 4.0 era (Bhatia, 2022). AI applications have become ubiquitous in the banking industry, significantly benefiting banks. The trend is vibrant in Malaysia, with a significant increase in financial service providers implementing AI and making it a strategic focus for the near future. GenAI has been the primary focus due to its great potential and the room for exploration and adoption. Thus, utilising AI technologies in banks is inevitable and is reshaping the entire banking industry.

1.2 Problem Statement

Banking 4.0 is the digital innovations that impact societies, industries, and firms with ever-increasing speed and intensity through the integration of the physical, artificial, and biological worlds. Following Fasnacht's (2021) discussion, conventional banking structures can be ill-equipped in the new digital era. However, customers often feel uncomfortable and hesitant when adopting innovations and new services. This resistance is particularly prominent for Banking 4.0, which promotes digital-only banks and eliminates most human-contact-based service systems (Nel & Boshoff, 2020, 2021). Additionally, Kim and Bae (2020) and Li et al. (2021) suggest that if customers perceive Banking 4.0 as inferior to traditional models in terms of security,

reliability, and quality, the adoption rates may decline. Therefore, studying the determinants that affect customers' adoption of Banking 4.0 technology is crucial.

In Malaysia, the banking industry is advancing towards the Banking 4.0 model. In 2022, Bank Negara Malaysia granted five digital bank licenses to enable individuals and firms to access more data-backed, personalised solutions (Bank Negara Malaysia, 2022). A nationwide KPMG survey (2020) found that 77% of Malaysians view digital banks as the future of financial services. Although 82% of respondents are current internet banking users, 57% of respondents are hesitant to bank with institutions that lack physical branches. This poses a significant challenge to a complete transition to Banking 4.0 in Malaysia. Failure to address these issues may lead Malaysian banks to fall behind the global digitalisation trend, be outcompeted by foreign competitors, lose customers, and face financial distress.

Noor and Mansor (2019) noted that Malaysia is advancing AI innovation to keep pace with the 4IR and National Transformation 2050 (TN50). The initiatives implemented by the Malaysian Digital Economy Corporation (MDEC) to support the National AI Framework ensure that Malaysia is developing an effective AI ecosystem (Abdullah et al., 2017). Zahid (2024) states that over 80% of banks globally have adopted AI-related projects, primarily to support client analytics, fraud detection, and electronic Know-Your-Customer (e-KYC) processes. Accenture reports that 70% of financial services executives in Malaysia believe AI and virtual assistants will be the leading players in the future of the banking industry's customer service (Hisham, 2024). KPMG (2024) reports that 70% of CEOs rank generative AI as a top investment priority. Ikhsan et al. (2024) state that 46% of global companies have adopted AI, but the adoption rate in Malaysia is only 15% to 20%. McKinsey & Co. found that AI adoption has enormous potential to improve Malaysia's productivity by eliminating 50% of repetitive work (MIDA, 2022). Kok and Siripipatthanakul (2023) argue that the relatively low adoption of AI in Malaysia is due to social challenges, including cultural norms, unrealistic expectations, insufficient knowledge, and a fear of job security. Therefore, studying these perception factors in AI adoption in Malaysia is essential for the successful implementation of AI banking and the Banking 4.0 transformation.

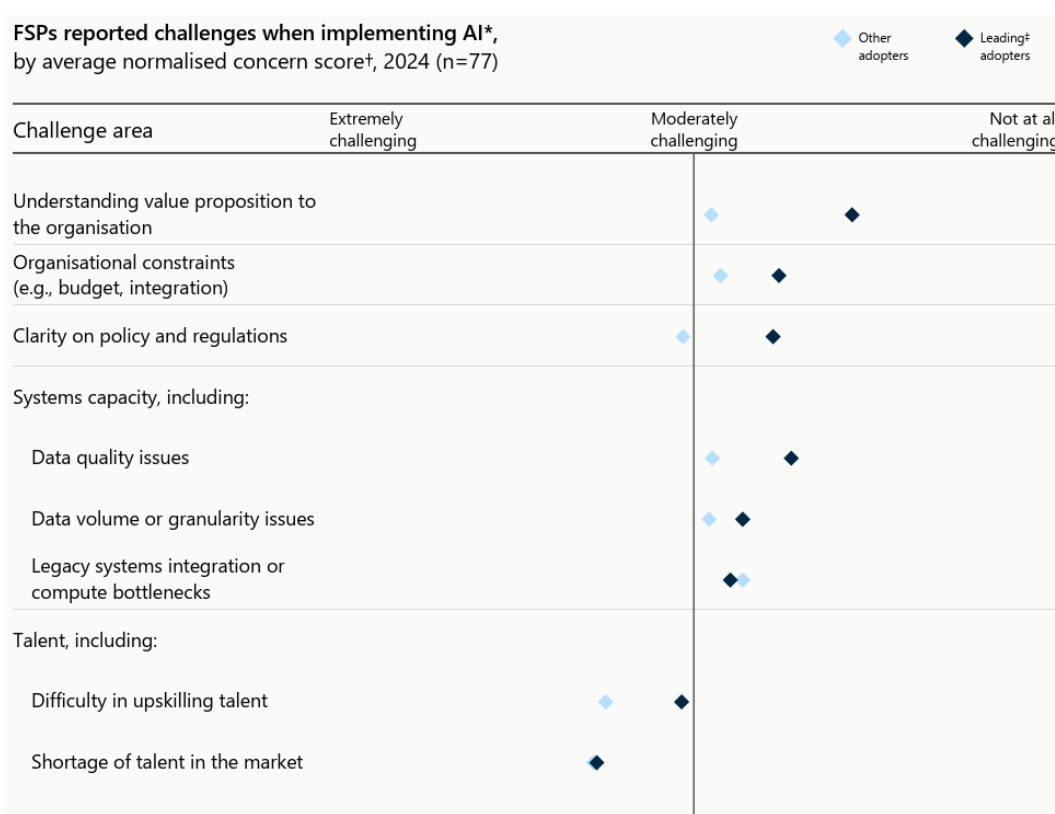
Generation Z comprises individuals born between 1996 and 2010 (McKinsey & Company, 2024), representing 29% of Malaysia's population. They are called the four 'E' generation because they are born to be electronically engaged, educated, entrepreneurial, and empowered (Tjiptono et al., 2020). These traits suggest that Gen Z may be more open to adopting technology and embracing digital transformation than any other generation. In their studies, Daqar et al. (2020) and Rosli et al. (2023) also highlighted Gen Z as the driving force behind global FinTech innovations and digital transformation. Despite Gen Z's natural advantage of greater exposure to and knowledge of technology, several factors influence their intention to adopt banking technology. The Technology Acceptance Model (TAM) proposes that perceptions are the key factors influencing variation in adoption intention among Gen Z. Therefore, scrutinising the perception elements that shape the adoption intention is essential to the success of the Banking 4.0 transformation in Malaysia.

Malaysia is shifting towards the Banking 4.0 model, and AI in banking is a fundamental part of the transformation plan. Although 70% of senior financial executives in Malaysia support the perspective that AI is the future of banking, the AI adoption rate remains relatively low in Malaysia, at 15% to 20%, which raises concerns. The most concerning fact is that approximately 47% of Gen Z in the U.S., one of the most developed economies, do not express interest in AI in banking (Pymnts, 2023). The contributing factors to such resistance are privacy, trust, and confidentiality concerns. Furthermore, Moden et al. (2023) state that Gen Z will comprise 27% of the workforce by 2025. They significantly influence the success of developing and implementing new banking technology. Besides, Gen Z is a racially and ethnically diverse generation. They are digital natives and globally conscious, which sets them apart from any other generation. Their behavioural intention may differ from that of other generations when adopting a particular technology. Given the limited research on Gen Z Malaysians' intentions to adopt AI in banking, it is necessary to conduct a study to understand Gen Z's priorities when adopting AI in banking.

According to Bank Negara Malaysia (2025), financial service providers in Malaysia identified several challenges in implementing AI within their institutions. The result indicates that talent shortages are the shared concern for both leading and other adopters (Figure 1.4). Consequently, Bank Negara Malaysia has outlined four key

areas for industry-wide collaboration to support more effective and responsible AI adoption across the financial sector. Firstly, developing industry guidelines and best practices for AI risk management and governance is vital. Besides, we must facilitate knowledge-sharing on AI implementation and risk management. Additionally, efforts should be taken to strengthen talent pipelines and AI expertise for the financial sector. Lastly, initiatives in enhancing consumers' financial and AI literacy are crucial. In alignment with these four industry-wide collaborations, the investigation into the public's perceptions of financial AI is critical for formulating a strategy to enhance AI adoption. Therefore, primary data research must be conducted to capture the most recent data to inform strategic initiatives.

Figure 1.4: Financial service providers' reported challenges when implementing AI



Source: Bank Negara Malaysia (2025)

The TAM has highlighted the significance of perceived usefulness and perceived ease of use in one's technology adoption intention. However, the original TAM omitted several factors that may influence Gen Z's adoption intention. Therefore, model extension is necessary to achieve more accurate predictions. Following Ashcraft (2023), Gen Z places less emphasis on privacy when considering adopting a

technology. This is due to their strong presence on social media, which makes much of their information publicly available. They are also sceptical about corporations' ability to safeguard their information, with only 39% trusting the companies. Shah (2025) identified the engagement gap that may explain Gen Z's non-adoption behaviour of AI in banking. He reveals that only 7% of Gen Z surveyed trust the advice of bank representatives, which makes them hesitant to utilise the new technology they propose. Trust's influence on Gen Z's adoption intention is suggested by Egilsdottir (2023), who indicates that Gen Z demands a transparent, value-aligned bank to boost their confidence as consumers. That said, the risk of privacy leakage and the trustworthiness of banks are essential determinants for Gen Z when deciding whether to adopt new banking technology. However, studies on risk and trust are limited. Therefore, additional research is required to understand how risk and trust affect the adoption intentions of Generation Z.

1.3 Significance of the Study

There are five key contributions of this research. Firstly, it examines the factors influencing Gen Z Malaysians' intention to adopt AI in banking services. This fills the research gap regarding generational differences in the adoption of AI banking technology in Malaysia. It also provides empirical evidence on whether the factors influencing Gen Z Malaysians' banking AI adoption intention are significantly different from those of Gen Zs from other nations. The findings serve as a benchmark for future cross-generational or cross-regional studies to compare differences among populations from different generations or cultural backgrounds. Thus, this research provides specific insights into generation and cross-country comparisons.

Secondly, it examines Gen Z's adoption intentions at a time when AI banking is still a relatively emerging phenomenon in Malaysia. As AI banking and the Banking 4.0 phenomenon become relevant and accessible, it is crucial to understand how adoption intentions evolve using timely data. The evolution can be used to predict future trends in banking AI adoption and to identify shifts in consumer preferences that require modifications to implementation strategies. These valuable insights enable policymakers and financial institutions to highlight the crucial perceptual factors that influence Gen Z's adoption intention. They utilise the insights to formulate strategies

that effectively increase Gen Z's adoption of AI banking and, ultimately, the adoption of Banking 4.0. Thus, this study captures changes in adoption intentions across time using the latest data collection to generate predictive and strategic value.

Thirdly, this study incorporates variables that have been shown to affect Gen Z's adoption intention in other contexts significantly. However, these factors remain underexplored empirically in the Malaysian Gen Z context. Therefore, this study examines whether the factors are valid in this context using localised data. The findings have crucial practical implications for banks and Fintech companies, as they guide banks in developing strategies to enhance the experience and loyalty of the largest group of future customers amid a drastic revolution. Thus, this research contributes by demonstrating the significance of the variables in the Gen Z Malaysian context and gauging their preferences to secure their support when implementing banking AI.

Fourthly, the findings provide empirical grounds for future researchers to hypothesise about the influence of perception factors on behavioural intention to adopt a particular technology. It offers a fundamental conceptual framework for future researchers who study technology acceptance behaviour. The predictive model can be incorporated directly or modified to test the relationships among variables across different topics. The conceptual framework is replicable in an emerging economy undergoing a Banking 4.0 transformation, enabling the study of the influence of perceptions on adoption intention in other similar economies. Hence, the research contributes a fundamental framework for future researchers when conducting related studies.

Lastly, this study has a significant societal impact, promoting financial inclusion among young people. The process and outcomes of this research can facilitate more effective and responsible adoption of AI in the financial industry. The distribution of the questionnaire exposed the topic to more Gen Zs, helping banks better reach younger generations, understand their adoption drivers, and prepare them for AI banking. Consequently, there will be a large population of a future-ready workforce for banks' AI development and operation. This skilled workforce can also be a significant driving force in the future evolution of the banking industry's technology. Thus, this research can facilitate knowledge-sharing on AI implementation and equip Gen Zs with a fundamental understanding of the emerging trend in banking AI.

1.4 Research Questions

In coherence with the above discussions, this study aims to resolve these questions:

1. What influence does perceived usefulness have on the intention of Malaysian Gen Z to adopt AI in banking?
2. What influence does the attitude towards AI in banking have on the intention of Malaysian Gen Z to adopt AI in banking?
3. What is the effect of perceived ease of use on the intention of Malaysian Gen Z to adopt AI in banking?
4. How does perceived risk impact Malaysian Gen Z in making banking AI adoption decisions?
5. How does perceived trust form Malaysian Gen Z's intention to adopt AI in banking?

1.5 Research Objectives

Throughout this study, several objectives are to be achieved:

1. Determine how perceived usefulness influences the intention of Gen Z Malaysians to adopt AI in banking.
2. Identify the effect of attitude towards AI in banking on the intention of Gen Z Malaysians to adopt AI in banking.
3. Investigate the impact of perceived ease of use on the adoption intention of AI in banking among Gen Z Malaysians.
4. Assess the influence of perceived trust on the intention of Gen Z Malaysians to adopt AI in banking.
5. Examine the effect of perceived risk on Gen Z Malaysians' intention to adopt AI in banking.

1.6 Organisation of the Study

The following parts of this study are organised. Chapter 2 outlines the relevant theories, conceptual framework, empirical evidence, and research gaps regarding AI banking adoption intention. Chapter 3 discusses the research design, hypothesis development, sample and population, data collection method, measurement of variables, proposed data analysis, and expected outcomes.

CHAPTER 2

LITERATURE REVIEW

2.0 Introduction

Chapter 2 introduces the underlying theoretical model, empirical findings, and research gaps from previous studies. Section 2.1 discusses the TAM. Section 2.2 summarises the empirical evidence from past studies. Finally, Section 2.3 identifies the research gaps that motivate this study.

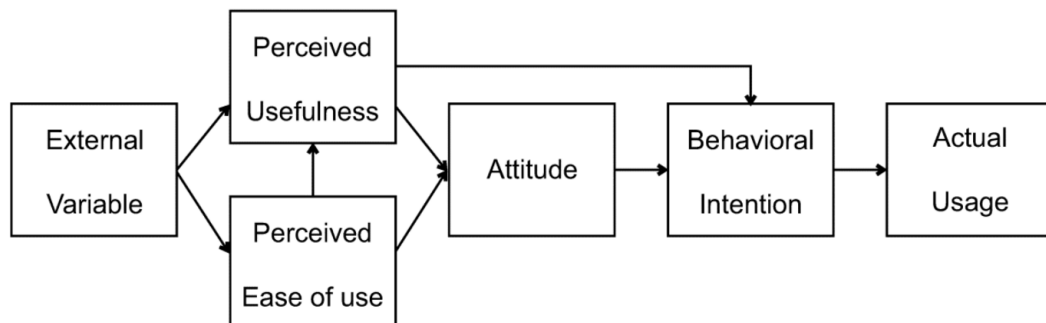
2.1 Theoretical Review

2.1.1 Technology Acceptance Model

Davis (1985) introduced TAM to assess user acceptance of a computer-based information system, accounting for its characteristics. Marangunić and Granić (2014) stated that TAM is a prominent model in explaining consumers' behaviour towards technology, and is rooted in two prestigious psychology-based models: TRA and TPB.

The model's initial assumption is that two variables—perceived usefulness and perceived ease of use—mediate the complicated relationship between potential system use and system characteristics. As defined by Davis (1989), perceived usefulness is the individual's perception of how a specific system enhances their productivity and work performance. Moreover, PEOU is one's belief that utilising a particular system will not burden them much or be effort-free. The fundamental relationships among variables in TAM are shown in Figure 2.1.

Figure 2.1: TAM model.



Source: Park and Park (2020)

Following Lee et al.'s (2003) statement, the TAM has evolved chronologically and continuously through four stages: introduction, validation, extension, and elaboration. During the introduction stage, researchers mainly focused on replicating TAM in other contexts and compared it to its origin, TRA, to examine whether it is superior. Studies in the validation period emphasise investigating the TAM's power, consistency, reliability, and validity. The findings confirmed all these properties in TAM (Szajna, 1994; Davis & Venkatesh, 1996). During the extension period, TAM studies emphasise the addition of external variables in the model. The effort to scrutinise and examine TAM's boundary conditions is another stream during this period. Lastly, the elaboration period focuses on developing a cutting-edge TAM that synthesises prior impacts and addresses constraints in earlier research.

Alshammari and Rosli (2020) argue that TAM is often applied due to its strength, soundness, and adaptability. It addresses TRA's limitations by including crucial perception factors. The extension flexibility also improves TAM's explanatory power in diverse applications and contexts. Given these advantages, this paper applies TAM to examine the AI banking adoption among Gen Z Malaysians.

2.1.2 Prospect Theory

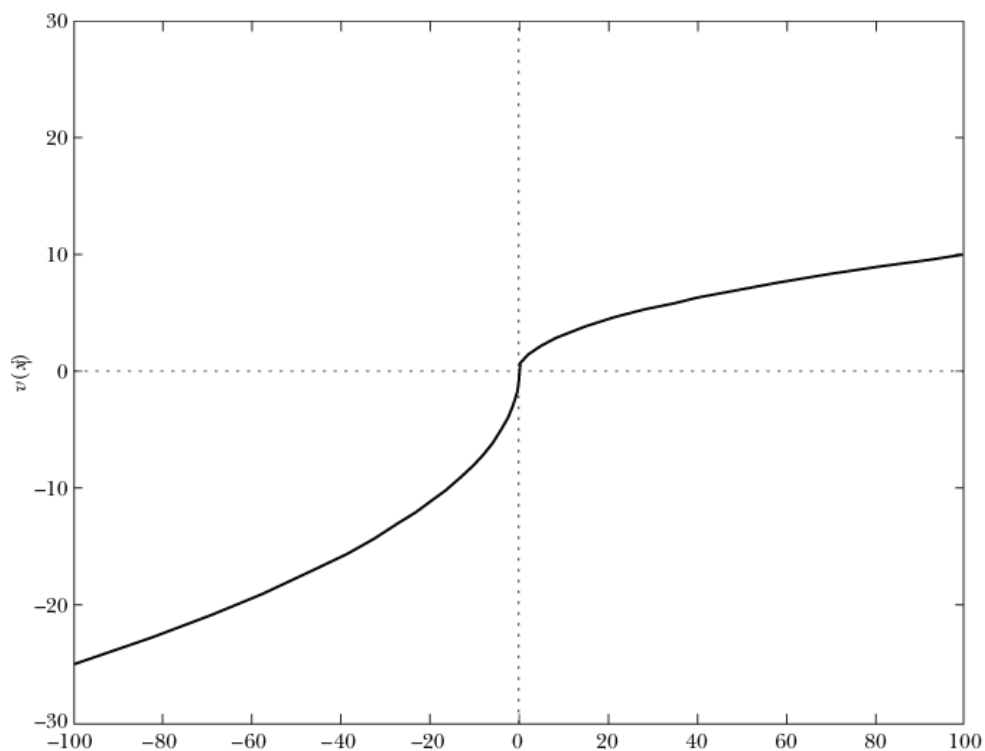
Kahneman and Tversky (1979) challenged the expected utility model by proposing the prospect theory to analyse risk-related decision-making. The prospect theory outlines editing and evaluation phases in the decision-making process. The editing phase involves the initial evaluation of the presented opportunities, while the evaluation phase evaluates and selects the highest-value prospects (Levy, 1992). The theory initially focused primarily on financial outcomes, but also mentioned its applicability to other decision-making situations involving different attributes, such as the technology adoption decision in our case.

Khan et al. (2022) utilised prospect theory to analyse the adoption of digital health technologies. The framing effect on decision-making during the editing phase is highlighted in their paper. For example, suppose the banking AI has improved the experience of one-third of clients. In that case, one can view it as either a 1/3 chance

that he will benefit from adopting the banking AI or a 2/3 chance that he will not. The difference in perspective significantly affects clients' adoption behaviour.

On the other hand, Barberis (2013) elaborates on four elements of prospect theory. Firstly, reference dependence occurs when individuals derive value from gains and losses by evaluating them against a reference point rather than the sheer amount of wealth when making choices. Secondly, loss aversion suggests that people focus more on losses than gains of identical magnitude, resulting in a non-linear value function, as illustrated in Figure 2.2. The horizontal axis denotes the dollar gain or loss, while the vertical axis represents the assigned value to the gain or loss. Thirdly, the value function is concave in the gain region but convex in the loss region. Consequently, people tend to be risk-averse when faced with moderate probability gains, but risk-seeking when faced with losses. Lastly, the theory posits that people weigh outcomes based on their transformed probabilities, or decision weights, rather than their objective probabilities. In such cases, individuals often place too much emphasis on low-probability events and give insufficient weight to high-probability ones.

Figure 2.2: The Prospect Theory Value Function.



Source: Francis (2024)

For our study, the reference point is Gen Z Malaysians' current banking experience without the involvement of banking AI. Therefore, according to prospect theory, their decision to adopt AI banking is based on the perceived change in value they can gain relative to this reference point, similar to a gain or loss in economic terms. Besides, the second core element, loss aversion, is powerfully relevant in our case. Gen Z Malaysians may perceive several potential losses, including financial and privacy losses. These risks will trigger their loss-avoidance behaviour, resulting in a lower intention to adopt AI in banking and a continued status quo. Hence, the interplay between perceived value and perceived risk must be analysed using prospect theory to examine banking AI adoption behaviour among Gen Z Malaysians.

Kahneman (2011) and Willemssen et al. (2011) highlight that people have an intrinsic preference for the current state of affairs over a possibly better alternative. Studies have shown that the perceived psychological loss should be doubled to convince an individual to shift away from their current state of affairs (Kahneman & Tversky, 1979; Kahneman, 2011; Tversky & Kahneman, 1992). Based on the theory, the potential gain from applying new technology must be twice the psychological loss incurred from implementing it to make the faculty willing to adopt it. In our case, the perceived benefit from adopting banking AI, such as time saving, 24/7 service, and personalised banking experience, should be perceived as twice the value of the perceived risk or loss to make Gen Z Malaysians willing to embrace this new technology.

In summary, prospect theory provides the foundational framework for incorporating perceived risk in our study. It explains that Gen Z Malaysian's banking AI adoption intention is more than a simple rational calculation. It is a highly subjective evaluation that can lead to a biased assessment, where one may perceive the gains as smaller and the losses as larger. Therefore, an empirical study is essential to accurately model the decision-making process of Gen Z Malaysians.

2.1.3 Trust

Moorman et al. (1992) define trust as the readiness to depend on a partner in an exchange whom one believes in. Alejandro and Güemes-Castorena (2024) have outlined empirical evidence demonstrating that trust influences users' behaviours

towards technology adoption (Chandra et al., 2010), usage continuance (Teo et al., 2008), and intention to use (Kim et al., 2009). As proposed in the TAM, critical determinants of adoption decision include perceived usefulness and perceived ease of use. However, individuals must make a ‘leap of faith’ under imperfect information to decide whether to adopt the new technology (Bahmanziari et al., 2003).

Lewis and Weigert (2012) suggest that trust is multidimensional. Nooteboom and Six (2003) support the argument by stating that trust is inherently complex, with distinct cognitive, emotional, and behavioural contexts. Bryant et al. (2025) summarise multidimensional trust into three typologies: cognitive, affective, and behavioural trust. Cognitive trust refers to the trustor's logical belief that the trustee possesses the qualities necessary for dependability (Komiak & Benbasat, 2006). It ensues when sufficient evidence supports logical reasoning and evaluation processes that build trust. According to Kaasinen (2005), cognitive trust can be evaluated based on a technology's accuracy, reliability, and privacy. In this study, it is Gen Z Malaysian's perception towards the accuracy, reliability, and privacy of the banking AI.

Cognitive trust is the rational aspect of trust formation, while affective trust is the emotional side. It can also be expressed as emotional, identity-based, relational, and affect-based trust (Akrouit et al., 2016; Chowdhury, 2005; Johnson & Grayson, 2005; Komiak & Benbasat, 2006). Komiak and Benbasat (2006) further elaborate emotional trust as the extent to which a person feels secure and comfortable in depending on the person they trust. Rempel et al. (1985) conclude that emotional trust is associated with a sense of safety and the perceived robustness of the relationship. Additionally, affective trust covers elements such as benevolence (Sekhon et al., 2014), confidence (McKnight & Chervany, 1996), and honest acts (Cummings & Bromiley, 1996). As AI in banking is still an emerging trend in Malaysia, affective trust will be an essential factor influencing perceived trust and the adoption intention.

The last aspect of trust is the behavioural trust, which is the application of cognitive and emotional trust (McKnight et al., 2002). Mayer et al. (1995) provided a more comprehensive definition to clarify the difference between the general concept of trust and behavioural trust. The difference is that trust concerns the “willingness” to assume risk, while trusting behaviours concern actually “assuming” the risk. In this sense,

Lewis and Weigert (1985) conclude that behavioural trust is the reciprocal dimension and realisable outcome of cognitive and emotional trust. Some of the behavioural trust measures are presenting available honest and transparent information (McKnight & Chervany, 1996), entering into informal agreements (Currall & Judge, 1995), risk-taking (Mayer et al., 1995), and willingness to make decisions based on the available information (Kim & Sundar, 2016). As AI banking is still in its early stages in Malaysia, behavioural trust among Gen Z Malaysians has yet to be measurable. Therefore, the study will primarily focus on measuring cognitive trust and affective trust.

2.2 Empirical Review

Studies have examined consumers' intention to adopt Banking 4.0 technology in Malaysia (Othman et al., 2019; Rahman et al., 2021; Said et al., 2022; Saif et al., 2024), but only a few focused on Gen Z (Rosli et al., 2023) or AI banking adoption (Rahman et al., 2021). Additionally, previous studies used traditional TAM variables, such as perceived usefulness and perceived ease of use (e.g., Rahman et al., 2021; Noreen et al., 2023; Sabir et al., 2023; Menezes et al., 2024). However, multiple studies have outlined the importance of perceived risk (e.g. Hu et al., 2019; Nawayseh, 2020; Hoang et al., 2021; Noreen et al., 2023; Menezes et al., 2024) and trust (e.g. Oney et al., 2017; Hu et al., 2019; Nawayseh, 2020; Said et al., 2022; Saif et al., 2022; Hurani & Abdel-Haq, 2025) in affecting consumers' Banking 4.0 adopting intention. This section summarises the related past literature and identifies the variables for this study.

2.2.1 Perceived Usefulness

Noreen et al. (2023) define perceived usefulness (PU) as the degree to which a particular system is believed to enhance performance. Numerous studies have included perceived usefulness when investigating intention to adopt a particular technology. However, the conclusions and approaches from their studies vary. Juwaheer et al. (2012), Ryu (2018), and Afshan et al. (2018) suggested that perceived usefulness positively influences intention to adopt. Hu et al. (2019) and Belanche et al. (2019) indicate that perceived usefulness influences adoption intention through attitude. However, the direct relationship is insignificant in Belanche et al.'s (2019) study. Yahaya et al. (2025) suggested that perceived usefulness is insignificant when

analysing Gen Y and Gen Z respondents separately, but significant when combined. Rahman et al. (2021), Sabir et al. (2023), and Menezes et al. (2024) propose that perceived usefulness has both direct and indirect influences on the intention to adopt, the latter being mediated by the attitude towards the technology. In contrast, two papers suggest only a positive relationship between perceived usefulness and adoption intention (Noreen et al., 2023; Singh et al., 2024). The divergent empirical findings provide an opportunity to investigate further the effect of perceived usefulness on the intention to adopt AI in banking.

H₁: Perceived Usefulness (PU) positively influences Intention to Adopt AI in Banking (INT).

2.2.2 Attitude towards AI Banking

Attitude (ATT) is frequently used in TAM as a vital mediator among the relationships that predict actual usage of a particular technology (Rosli et al., 2023). The position of attitude varies across models. Some suggest that a direct effect of attitude towards a specific technology on intention to adopt exists, while others argue that it is a mediating variable. Juwaheer et al. (2012), Aggarwal et al. (2023), Noreen et al. (2023), and Singh et al. (2024) suggest that attitude has a positive, direct influence on the intention to adopt. Besides, attitude is a significant factor influencing intention to adopt and also serves as a mediator for PU (Belanche et al., 2019; Hu et al., 2019; Hoang et al., 2021). Following Rahman et al. (2021), attitude is a significant mediator between perceived usefulness and intention to adopt. Alshari and Lokhande (2022) found a strong positive relationship between attitude and intention to adopt, and identified mediating roles for attitude in the relationships between perceived usefulness and intention to adopt and between perceived ease of use and intention to adopt. This conclusion is supported by Rosli et al. (2023) and Sabir et al. (2023). Menezes et al. (2024) argue that although perceived usefulness and perceived ease of use positively affect attitude, attitude does not directly affect intention to adopt. The role of attitude in predicting intention varies; therefore, examining its influence is essential.

H₂: Perceived Usefulness (PU) positively influences Intention to Adopt AI in Banking (INT) through Attitude (ATT).

2.2.3 Perceived Ease of Use

Kumar and Dami (2021) define PEOU as the belief that equipping oneself with the skills to master a technology requires minimal effort. Perceived ease of use (PEOU) is the key variable in the TAM that is widely incorporated when examining intention to adopt. However, the empirical results in these studies vary from case to case. Juwaheer et al. (2012), Afshan et al. (2018), Othman et al. (2019), and Kumar and Dami's (2021) identify a positive relationship between perceived ease of use and intention to adopt. Numerous studies suggest that perceived ease of use positively influences the intention to adopt, either through perceived usefulness (Hu et al., 2019; Singh et al., 2020; Solarz & Adamek, 2021) or through both perceived usefulness and attitude (Belanche et al., 2019; Sabir et al., 2023). In contrast, Hoang et al. (2021), Rahman et al. (2021), and Hurani and Abdel-Haq (2025) argue that perceived ease of use has no statistically significant influence on intention. Yahaya et al. (2025) report a mixed result: Perceived Ease of Use has a significant influence on intention among Gen Zs but not among Gen Ys, and it is also significant when analysing Gen Y and Gen Z together. Menezes et al. (2024) proposed both direct and indirect relationships between perceived ease of use and intention to adopt. The indirect relationship involves attitude towards AI in banking as the mediator. To sum up, researchers suggest that a direct, an indirect, or both relationships exist. Given the inconsistent findings, this study assesses the relationship between perceived ease of use and intention to adopt AI in banking.

H₃: Perceived Ease of Use (PEOU) positively influences Intention to Adopt AI in Banking (INT) through Attitude (ATT).

H₄: Perceived Ease of Use (PEOU) positively influences Intention to Adopt AI in Banking (INT) through Perceived Usefulness (PU).

2.2.4 Perceived Risk

As Bauer (1967) defined, perceived risk (PR) combines uncertainty and the severity of the outcome. It is a negative factor that discourages consumers from adopting a particular financial service or technology (Li et al., 2016). Ryu (2017), Afshan et al. (2018), Rahman et al. (2021), Noreen et al. (2023), Menezes et al. (2024), and Salmasi et al. (2024) suggest that perceived risk negatively influences intention to adopt. In

contrast, Nawayseh (2020), Hoang et al. (2021), Lee and Chen (2022), and Saif et al. (2022) reject the notion of a direct relationship. Other studies reveal the indirect influence of perceived risk on intention to adopt through trust (Nawayseh, 2020), attitude (Alshari & Lokhande, 2022), or both (Hu et al., 2019). Additionally, Solarz and Adamek (2021) suggest that perceived risk is directly and indirectly related to the intention to adopt, with perceived ease of use and behavioural intention mediating these relationships. Saif et al. (2024) reran an identical study two years later and found that functional risk influences intention to adopt through perceived value, which fully mediates the relationship between the two. Yahaya et al. (2024) found that perceived risk, as measured by perceived data security and privacy, is an insignificant factor influencing the adoption of e-banking among Gen Z Malaysians. However, the result is significant among Gen Y Malaysians and when analysing both generations together. Given the mixed findings, further investigation into the relationship between perceived risk and the intention to adopt AI in banking is crucial to clarify the influence.

H₅: Perceived Risk (PR) negatively influences Intention to Adopt AI in Banking (INT) through Attitude (ATT).

2.2.5 Perceived Trust

Mayer et al. (1995) define trust as one's readiness to be exposed due to others' actions, anticipating that others will perform a particular action in favour of them, regardless of the capability to screen or manage the others. Trust is crucial in social interactions, as it alleviates an individual's worries and enhances commitment. Perceived trust (PT) is found to impact Internet banking adoption directly in multiple studies (Hoo et al., 2024; Juwaheer et al., 2012; Oney et al., 2017; Afshan et al., 2018; Othman et al., 2019; Rahman et al., 2021; Said et al., 2022; Saif et al., 2022). Following Hu et al. (2019) and Nawayseh (2020), perceived trust mediates the relationship between perceived risk and the intention to adopt. Hoang et al. (2021) and Alshari and Lokhande (2022) conclude that perceived trust positively influences the intention to adopt through attitude. Additionally, Hurani and Abdel-Haq (2025) suggest that perceived trust positively affects Fintech adoption, with perceived usefulness and perceived ease of use as mediators. In contrast, Salmasi et al. (2024) argue that perceived trust does not impact intention to adopt. To reach an accurate conclusion for Gen Z Malaysians, this

study analyses the relationship between perceived trust and the intention to adopt AI in banking.

H₆: Perceived Trust (PT) positively influences Intention to Adopt AI in Banking (INT) through Attitude (ATT).

2.3 Research Gap

Existing studies have studied the intention to adopt Banking 4.0 technology. However, Banking 4.0 encompasses a broad scope of cutting-edge technologies in the financial industry, including internet banking (Juwaheer et al., 2012; Afshan et al., 2018; Othman et al., 2019) and digital-only banks (Saif et al., 2022; Saif et al., 2024; Shaikh & Amin, 2024). However, studies focusing on the adoption of AI in banking are limited (Rahman et al., 2021; Noreen et al., 2023; Sabir et al., 2023; Menezes et al., 2024; Singh et al., 2024). Thus, it is crucial to narrow the focus to AI in banking to gauge consumers' adoption intentions more precisely.

Although numerous studies have examined consumers' perspectives on the emerging Banking 4.0 revolution, their geographical coverage varies. There are studies covering consumers from economies including China (Hu et al., 2019; Sabir et al., 2023), India (Bhatt & Bhatt, 2016; Singh et al., 2020; Aggarwal et al., 2023; Menezes et al., 2024), Indonesia (Kumar & Dami, 2021), Mauritius (Juwaheer et al., 2012), Pakistan (Afshan et al., 2018), Poland (Solarz & Adamek, 2021), and Vietnam (Hoang et al., 2021). However, studies covering only Malaysians are minimal (Othman et al., 2019; Rahman et al., 2021; Said et al., 2022; Saif et al., 2022; Rosli et al., 2023). Due to differences in culture, economic conditions, technology literacy, and AI adoption rates, findings across countries vary significantly. Therefore, a Malaysian-based study should be conducted to address this research gap and lay the foundation for cross-country comparisons.

Additionally, past empirical studies have covered a broad age group. Only a few studies have specified the study population as Gen Y (Kumar & Dami, 2021; Aggarwal et al., 2023) or Gen Z (Rosli et al., 2023). Saif et al. (2022) and Aggarwal et al. (2023) have highlighted limitations in their studies that exclude the Gen Z population. They suggest that future research on Gen Z is essential and that the findings should be

compared with past findings. Hence, this paper focuses solely on Gen Z Malaysians' intention to adopt AI in banking. On the other hand, the dynamic behaviour of consumers has created a demand for empirical results based on the latest data to yield more timely findings and enable comparison. The existing literature lacks studies using the most recent data, underscoring the need to conduct research utilising 2025 data.

The TAM is widely used to predict an individual's intention to accept technology. Many studies have incorporated the core variables proposed in the TAM (Juwaheer et al., 2012; Afshan et al., 2018; Hu et al., 2019; Othman et al., 2019; Singh et al., 2020; Hoang et al., 2021; Kumar & Dami, 2021; Rahman et al., 2021; Solarz & Adamek, 2021; Alshari & Lokhande, 2022; Noreen et al., 2023; Rosli et al., 2023; Sabir et al., 2023; Menezes et al., 2024; Hurani & Abdel-Haq, 2025). However, the external variables incorporated changes from case to case. Although PR and PT are significant factors affecting adoption intention in many studies, the literature is scant in extending the TAM to incorporate them (Afshan et al., 2018; Hu et al., 2019; Hoang et al., 2021; Rahman et al., 2021; Alshari & Lokhande, 2022). Thus, the extended TAM, incorporating these two variables, must be tested to examine their impact on Gen Z Malaysians' intention to adopt technology.

2.4 Chapter Summary

Section 2.1 discusses the basics of the TAM, its development, and practical applications. Section 2.2 analyses empirical findings from prior literature to serve as the basis for the study, which examines the AI banking adoption intention of Gen Z Malaysians using five variables. Section 2.3 identifies four research gaps.

CHAPTER 3

RESEARCH METHODOLOGY

3.0 Introduction

This chapter introduces the research methodology, design, conceptual framework, variables, questionnaire setup, and expected outcomes. Sections 3.1 and 3.2 discuss the research design and the conceptual framework. Section 3.3 lists the hypothesis development and the equation modelling. Section 3.4 outlines variable measurement and questionnaire design. Section 3.5 showcases the proposed data analysis method, and Section 3.6 discusses the pilot study.

3.1 Research Design

3.1.1 Quantitative Research

Qualitative research entails gathering and examining data that is not numerical. It is often conducted to understand unquantifiable ideas, opinions, and experiences through methods such as interviews and surveys (Bhandari, 2020). Ghanad (2023) explained that qualitative research aims to identify the reasons for actions and individual perception. It generates general understanding towards a topic and formulates ideas and hypotheses.

In contrast, quantitative research entails the collection and analysis of numerical data to quantify the sampling data and generalise findings from multiple perspectives. Such research requires a series of procedures, from collecting, analysing, and interpreting quantitative data in a logical manner to examining theories or hypotheses (Bryman, 2016).

The objective of this research is to identify the factors that affect Gen Z Malaysians' intention to adopt AI in banking, using hypothesis testing. Since the data will be gathered on a quantified Likert scale, a quantitative research design is more appropriate. The study will rely on primary data from Gen Z Malaysians to analyse the factors influencing their intention to embrace AI in banking. To gather information on their behavioural intentions, a carefully designed questionnaire will be distributed to

the target group. The data collected is cross-sectional, capturing Gen Z Malaysians' behavioural intention to adopt AI in banking at the time they completed the questionnaire. This study will incorporate five independent variables (PU, PEOU, ATT, PR, and PT) to test their influence on INT. In addition to testing the direct relationship, the mediating roles of PU, ATT, and PT will be examined.

3.1.2 Population and Sample

Specifically, Gen Z refers to the population born between 1996 and 2010 (McKinsey & Company, 2024). In 2025, Gen Z spans ages 15 to 29, primarily comprising secondary and tertiary students, as well as young professionals and employees. The population of this study is Gen Z Malaysians. Therefore, this study will exclude non-citizens residing in Malaysia due to cultural differences. According to the Department of Statistics Malaysia (n.d.), as of 1 January 2025, Gen Z Malaysians accounted for approximately 7.6789 million out of 30.851 million Malaysian citizens, or 24.89% of the Malaysian citizens' population (Government of Malaysia, n.d.). This study will collect data on the tertiary student group to represent Gen Z's overall intention to adopt AI in banking. Due to time constraints, a convenience sampling technique will be employed. Therefore, the target respondents are students at colleges and universities in Malaysia.

Hassan et al. (2006) characterised a pilot study as a preliminary trial of research methods, data-collection tools, data collection strategies, and other research techniques before conducting the larger study. It helps identify shortcomings and flaws in research tools and procedures before running the full study (Lancaster et al., 2004). Additionally, it helps determine whether the study method, in our case, a questionnaire, is adequate for the study. Upon passing the pilot test, the survey questionnaire will be shared via email, social media, and participants' networks.

Krejcie and Morgan (1970) proposed that the adequacy of a research sample can be assessed using the following formula, and the results are summarised in Table 3.1.

$$s = \chi^2 NP(1-P) / d^2(N-1) + \chi^2 P(1-P)$$

or

$$s = \frac{\chi^2 NP(1-P)}{d^2(N-1) + \chi^2 P(1-P)}$$

Where s = required sample size

χ^2 = the table value of chi-square for 1 degree of freedom at the desired confidence level of .05= (3.841)

N = the population size

P = the population proportion (assumed to be .05 since this would provide maximum sample size)

d = the degree of accuracy expressed as a proportion (.05).

Table 3.1: Krejcie and Morgan Sample Size Determination Table

<i>N</i>	<i>S</i>	<i>N</i>	<i>S</i>	<i>N</i>	<i>S</i>
10	10	220	140	1200	291
15	14	230	144	1300	297
20	19	240	148	1400	302
25	24	250	152	1500	306
30	28	260	155	1600	310
35	32	270	159	1700	313
40	36	280	162	1800	317
45	40	290	165	1900	320
50	44	300	169	2000	322
55	48	320	175	2200	327
60	52	340	181	2400	331
65	56	360	186	2600	335
70	59	380	191	2800	338
75	63	400	196	3000	341
80	66	420	201	3500	346
85	70	440	205	4000	351
90	73	460	210	4500	354
95	76	480	214	5000	357
100	80	500	217	6000	361
110	86	550	226	7000	364
120	92	600	234	8000	367
130	97	650	242	9000	368
140	103	700	248	10000	370
150	108	750	254	15000	375
160	113	800	260	20000	377
170	118	850	265	30000	379
180	123	900	269	40000	380
190	127	950	274	50000	381
200	132	1000	278	75000	382
210	136	1100	285	100000	384

Note.—*N* is population size. *S* is sample size.

Source: Krejcie & Morgan (1970)

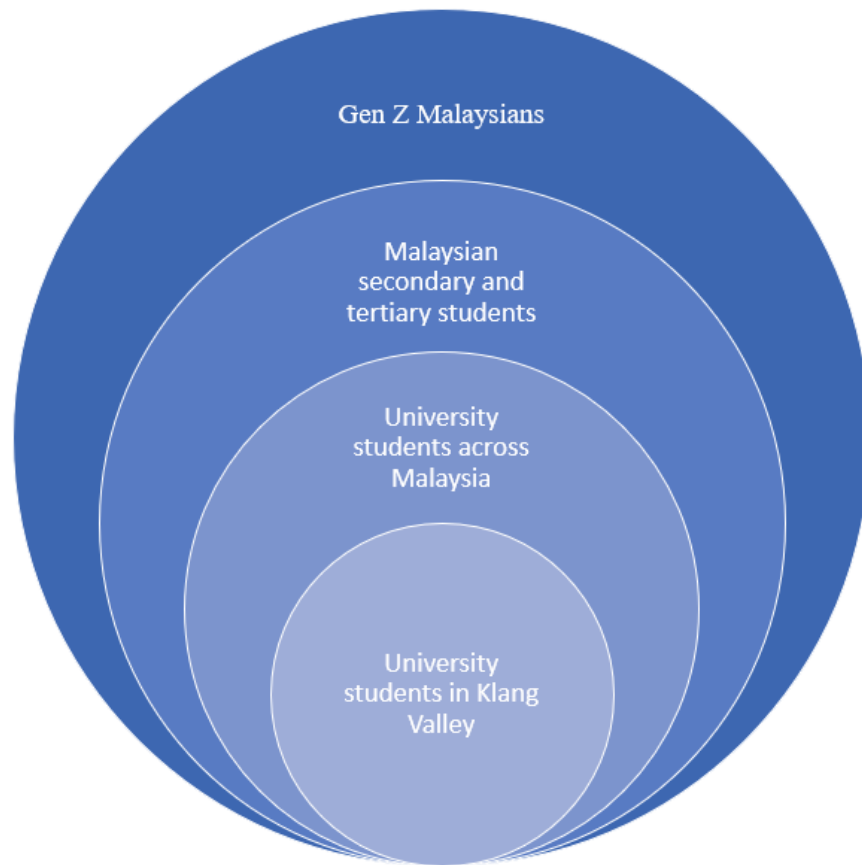
As the population of Gen Z Malaysians is approximately 7.6789 million, the adequate sample size, according to the table, is 384. Therefore, this study will collect data from at least 384 respondents to improve the generalisability of the findings. Specific initiatives are taken to address the non-response bias. Firstly, the survey is concise and straightforward, allowing respondents to complete it quickly and thereby enhancing their willingness to respond. Secondly, the questionnaire will inform potential respondents about what they can expect from the survey, including the estimated time required to complete it. Thirdly, the questionnaire will include an assurance of confidentiality for the collected data to ensure compliance with research ethics. Fourthly, a reminder will be sent to those non-respondents before the data collection deadline.

3.1.3 Sampling Method

Sampling methods are classified into two categories. Firstly, probability sampling ensures that every subject in the target population has an equal chance of being selected. Secondly, non-probability sampling entails unequal chances that each element in the population will be selected (Elfil & Negida, 2017). As the complete list of the target population is unavailable, it is impossible to ensure equal selection chances for inclusion. Therefore, non-probability sampling will be employed.

Specifically, convenience sampling is used due to time constraints. The sample selection depends on the researcher's ease of access and proximity until the desired sampling size is reached. Figure 3.2 shows the sampling frame. As the researcher is a university student studying in the Klang Valley, the most accessible group of subjects is the university peers in the Klang Valley. The second most accessible group is tertiary students from universities across Malaysia. Subsequently, the third group consists of Malaysian secondary school and tertiary students studying abroad. Lastly, the Gen Z Malaysian population comprises both employed and unemployed non-student Gen Zs. The questionnaires are delivered to the subjects via social media platforms, including WhatsApp, WeChat, Instagram, LinkedIn, and Microsoft Teams, as well as email. All respondents participate in the survey voluntarily and anonymously.

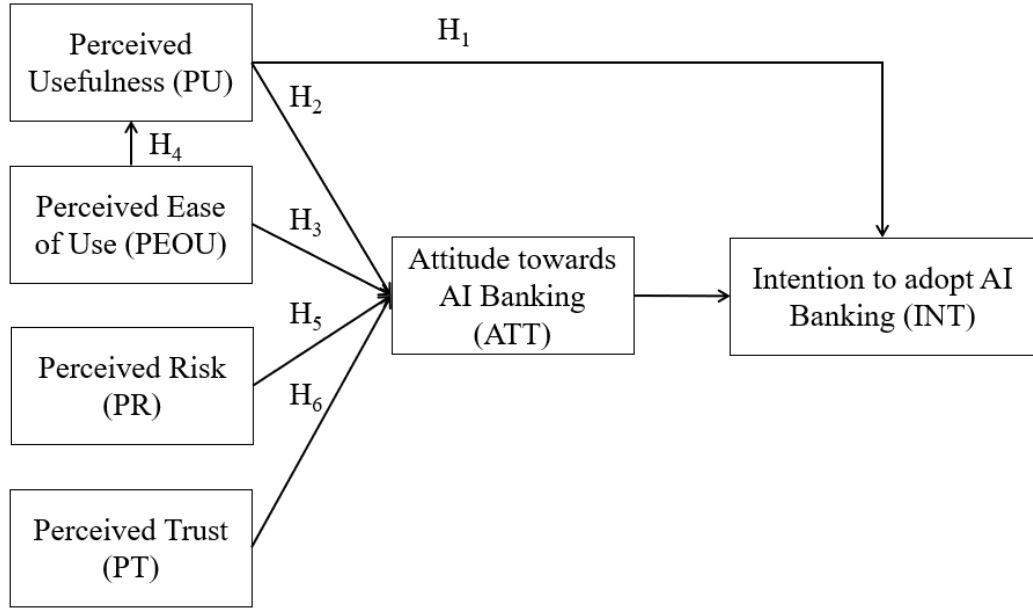
Figure 3.2: Sampling frame



3.2 Conceptual Framework

The conceptual framework is displayed in Figure 3.3. It is constructed according to Alshari and Lokhande's (2022) extended TAM with external variables, incorporating perceived risk and perceived trust.

Figure 3.3: Conceptual framework



3.3 Hypothesis Development

3.3.1 Hypothesis Summary

Table 3.4: Hypothesis summary

	<i>Hypothesis</i>	<i>Expected Path Relationships</i>	<i>Expected Net Effect</i>
H ₁	PU → INT	PU → INT (+)	Positive
H ₂	PU → ATT → INT	PU → ATT (+) → INT (+)	Positive
H ₃	PEOU → ATT → INT	PEOU → ATT (+) → INT (+)	Positive
H ₄	PEOU → PU → INT	PEOU → PU (+) → INT (+)	Positive
H ₅	PR → ATT → INT	PR → ATT (-) → INT (-)	Negative
H ₆	PT → ATT → INT	PT → ATT (+) → INT (+)	Positive

3.3.2 Model Equation

$$INT = \beta_0 + \beta_1 PU + \beta_2 ATT + \beta_3 PEOU + \beta_4 PT + \beta_5 PR + \epsilon$$

3.4 Measurement of Variables

The data on perception factors measuring Gen Z Malaysians' intention to adopt AI in banking were collected on a 5-point Likert scale. A higher integer value on the scale reflects respondents' agreement with the provided statement, and vice versa. A few questions will be set up for each variable to gauge the precise data regarding Gen Z Malaysians' intention towards AI in banking. The first part of the questionnaire collects demographic information, and the second part measures variables. Data on demographic factors will be used for descriptive analysis, and the average Likert-scale score for each variable will be used to perform the analysis. The questionnaire layout is presented in Table 3.5, and the questionnaire question construct is shown in Appendix 3.1.

Table 3.5: Questionnaire layout summary

<i>Variable</i>	<i>No. of questions</i>	<i>Question Code</i>
Demographic Factors	5	D1 – D5
Perceived Usefulness	4	PU1 – PU4
Perceived Ease of Use	4	PEOU1 – PEOU4
Attitude	4	ATT1 – ATT4
Perceived Risk	4	PR1 – PR4
Perceived Trust	4	PT1 – PT4
Intention to Adopt AI in Banking	3	INT1 – INT3
Total	28	

3.5 Data Analysis

3.5.1 Reliability and Validity Test

Following Williams et al. (2010), factor analysis is commonly applied in psychology and education to interpret self-report questionnaires. Three crucial functions of factor analysis are elaborated: firstly, it minimises the number of variables. Secondly, it defined the core links between observed variables and hidden constructs, thereby developing and strengthening the theory. Thirdly, it demonstrates the construct validity of the self-reporting scales. Thus, factor analysis is critical for examining data validity.

Explanatory factor analysis (EFA) and confirmatory factor analysis (CFA) are two major types of factor analysis. EFA is a multivariate statistical method for validating and constructing psychological theories (Watkins, 2018). It is used to justify the covariation between the measured variables by identifying the fewest hypothetical constructs that account for the observed relationship.

CFA is another sophisticated statistical technique that verifies whether the observed variables' factors and loadings align with the underlying theory (Malhotra et al., 2007). Sarmiento and Costa (2019) state that conducting reliability and validity tests is crucial for confirming and accurately representing the constructs observed in the variables. Hence, this study will perform reliability and validity tests before proceeding to hypothesis testing.

3.5.2 Hypothesis Test

Structural Equation Modelling (SEM) is described by Hair et al. (2021) as an advanced multivariate analysis method for evaluating complex relationships among constructs and their indicators. It surpasses the limitations of first-generation techniques by enabling the modelling and estimation of intricate relationships among variables. Thus, SEM will be applied in this research to estimate path relationships.

Two popular methods in SEM are Covariance-Based Structural Equation Modelling (CB-SEM) and Partial Least Squares-Structural Equation Modelling (PLS-SEM). Hair et al. (2021) noted that CB-SEM is primarily employed to validate theories and core assumptions. In contrast, PLS-SEM is both causal and predictive, while accounting for fluctuations in the dependent variables (Chin et al., 2020). PLS-SEM will be employed in this study due to its characteristics. Firstly, it can deal with both single-item and multi-item measured constructs. Secondly, it can easily incorporate reflective and formative measurement models. Thirdly, it can handle multiple relationships among structural models. Given the objective of predicting intention using perception factors, PLS-SEM should be preferred (Hair et al., 2017; Latan & Noonan, 2017; Hair et al., 2019; Shmueli et al., 2019).

Using PLS-SEM to test a theory requires two steps. Firstly, we must verify the measurement model's reliability and validity. Subsequently, we test the structural

theory (Hair et al., 2021). Hair and Alamer (2022) state that reliability and validity are assessed via five steps. The first step is to estimate the loadings and their p-values. The second step is estimating indicator reliability. Followed by examining internal consistency reliability. The fourth step is to obtain the Average Variance Extracted. Lastly, use the Heterotrait-Monotrait (HTMT) ratio to check the discriminant validity. After confirming the reliability and validity, four steps are taken to assess the structural model. Firstly, we detect the collinearity in the model. Secondly, we consider the paths' size and significance. Thirdly, we evaluate the R-squared. Lastly, we investigate the model's predictive ability on data not included in the sample.

In summary, this study will employ PLS-SEM using the two-step approach to test the hypothesis and verify the relationships among the variables. Additionally, the mediation analysis will be conducted to examine whether there are partial or complete mediation relationships in the model.

3.6 Pilot Study

Simkus (2023) described a pilot study as a preliminary, small-scale investigation conducted before the primary research to determine the adequacy of the research design and pinpoint aspects that may need improvement. The primary purpose is to identify and avoid major flaws in the study before conducting a large-scale study; thus, preventing the waste of time and money that would occur if the study had to be rerun due to a significant flaw (Polit & Beck, 2017). Additionally, a pilot test can assess the questionnaire's reliability according to the threshold using specific instruments.

Cronbach's alpha is used to assess the questionnaire's construct reliability. It evaluates internal consistency to determine whether multiple items belong to the same group. The satisfactory reliability threshold is set at exceeding 0.7, as indicated in Appendix 3.2. Using the 34 samples collected for the pilot testing, Cronbach's alpha for each item was calculated in SPSS and summarised in Table 3.6. As all Cronbach's alphas exceed 0.7, the questionnaire and its variables are internally consistent. Therefore, full-scale research can be executed.

Table 3.6: Pilot study's Cronbach's alpha

<i>Items</i>	<i>Cronbach's Alpha</i>
All variables	0.797
Perceived Usefulness	0.811
Perceived Ease of Use	0.821
Attitude Towards AI in Banking	0.776
Perceived Risk	0.840
Perceived Trust	0.852
Intention to Adopt AI in Banking	0.892

3.7 Chapter Summary

This is a primary, quantitative research that collects data via a questionnaire using convenience sampling. The study population is 7.6789 million Gen Z Malaysians, with an adequate minimum sample size of 384. This study involves testing a total of six hypotheses. The questionnaire collects data on demographic factors and measures variables on a 5-point Likert scale via 28 questions. CFA is employed to test the construct's reliability and validity, while PLS-SEM is used for hypothesis testing. The initial pilot test results indicate that the research design is internally consistent and can proceed to the large-scale data collection and analysis.

CHAPTER 4

RESULTS AND FINDINGS

4.0 Introduction

This chapter introduces the demographic analysis, reliability and convergent validity analysis, and structural model analysis. Sections 4.1 discuss the respondent demographics. Section 4.2 summarises the results of the reliability and validity tests. Section 4.3 analyses the structural model and hypothesis testing results.

4.1 Demographic Analysis

The demographics of 633 respondents collected from 12 September to 19 October 2025 are summarised in Table 4.1. Among the respondents, males comprise 44.87%, while females comprise 55.13%. Most respondents are aged 18–21, accounting for 53.71% of the total. The second is 22 to 25 years old, at 39.18%, followed by 26 to 29 years old at 5.85%, and 14 to 17 years old at 1.26%. The respondents are dominated by the students, who account for 75.20% of the total. Employed respondents account for 19.75%, while the unemployed account for 5.06%. Additionally, 87.36% of respondents hold or are pursuing a bachelor's degree. 4.74% is recorded for master's degrees, 3.48% for diplomas and equivalents, 2.37% for high school, 1.26% for other, and 0.79% for doctoral degrees. As for AI usage per week, 30.33% of respondents used AI 6 to 10 times, 25.43% used it more than 20 times, 18.48% used it 11 to 15 times, 17.38% used it 0 to 5 times, and 8.37% used it 16 to 20 times.

Table 4.1: Demographic profile

<i>Demographics</i>	<i>Categories</i>	<i>Frequency</i>	<i>Percentage</i>
Gender	Male	284	44.87
	Female	349	55.13
Age	14 - 17	8	1.26
	18 – 21	340	53.71
	22 – 25	248	39.18
	26 – 29	37	5.85
Employ Status	Employed	125	19.75
	Unemployed	32	5.06
	Student	476	75.20
Education	High School	15	2.37
	Bachelor Degree	553	87.36
	Master Degree	30	4.74
	Doctoral Degree	5	0.79
	Diploma or Equivalent	22	3.48
	Other	8	1.26
AI Usage Per Week	0 – 5	110	17.38
	6 – 10	192	30.33
	11 – 15	117	18.48
	16 – 20	53	8.37
	> 20	161	25.43

Source: Author's computation

4.2 Reliability and Validity Analysis

The latent-variable correlation matrix is a recommended method for assessing common-method bias (Bagozzi & Yi, 1990). The threshold is that the correlation value should not exceed 0.90 to verify the lack of common method bias. Table 4.2 demonstrates that no correlation value exceeds 0.9, with values ranging from -0.176 to 0.763. The values are well below the 0.9 threshold. Therefore, the study is free from common-method bias.

Table 4.2: Latent variable correlation

	ATT	INT	PEOU	PR	PT	PU
ATT	1.000					
INT	0.763	1.000				
PEOU	0.577	0.549	1.000			
PR	-0.118	-0.172	0.070	1.000		
PT	0.635	0.719	0.502	-0.176	1.000	
PU	0.716	0.635	0.617	-0.016	0.540	1.000

Source: Author's computation

This study follows several standard criteria proposed by Hair et al. (2014) to analyse the model's reliability and validity. The threshold for Cronbach's alpha (CA) and Composite Reliability (CR) is that the value should exceed 0.7 but not more than 0.95. As summarised in Table 4.3, the CAs ranged from 0.85 to 0.91, and the CRs ranged from 0.89 to 0.94, meeting the threshold. Therefore, we conclude that the constructs' reliability is satisfactory.

On the other hand, the factor loadings should exceed 0.708, and the Average Variance Extracted (AVE) should exceed 0.50 to confirm convergent validity (Sarstedt et al., 2017). The factor loadings are well above 0.708, with values ranging from 0.777 to 0.920. The AVEs also meet the threshold, with values ranging from 0.688 to 0.835. Thus, the model's convergent validity is confirmed.

Table 4.3: Measurement model's reliability and convergent validity

<i>Constructs</i>	<i>Items</i>	<i>Loadings</i>	<i>Mean</i>	<i>Std. Dev</i>	<i>CA</i>	<i>CR</i>	<i>AVE</i>
Perceived Usefulness	PU1	0.852	3.581	0.959	0.885	0.921	0.744
	PU2	0.856	4.055	0.863			
	PU3	0.856	4.112	0.825			
	PU4	0.886	3.877	0.949			
Perceived Ease of Use	PEOU1	0.811	3.828	0.920	0.850	0.899	0.691
	PEOU2	0.866	3.773	0.960			
	PEOU3	0.868	3.684	0.968			
	PEOU4	0.777	3.711	0.988			
Attitude Towards AI in Banking	ATT1	0.859	3.812	1.020	0.868	0.910	0.716
	ATT2	0.818	3.795	0.937			
	ATT3	0.865	3.798	0.995			
	ATT4	0.842	3.853	1.047			
Perceived Risk	PR1	0.833	3.367	1.116	0.852	0.898	0.688
	PR2	0.818	3.667	1.040			
	PR3	0.820	3.622	1.039			
	PR4	0.848	3.610	1.065			
Perceived Trust	PT1	0.882	3.185	0.950	0.879	0.917	0.734
	PT2	0.827	3.466	0.976			
	PT3	0.872	3.204	0.998			
	PT4	0.844	3.006	1.127			
Intention to Adopt AI in Banking	INT1	0.916	3.575	1.003	0.901	0.938	0.835
	INT2	0.905	3.735	0.960			
	INT3	0.920	3.577	1.029			

Source: Author's computation

Discriminant validity was tested using the Fornell-Larcker (FL) criterion and the Heterotrait-Monotrait (HTMT) ratio. The FL criterion of the latent constructs is exhibited in Table 4.4. According to Fornell and Larcker (1981), discriminant validity is supported if the diagonal values exceed the inter-construct correlations. Table 4.4 illustrates that the diagonal values range from 0.830 to 0.914, and the off-diagonal values range from -0.176 to 0.763. Since the lowest diagonal value is above the highest inter-construct correlation, discriminant validity is established.

Table 4.4: Fornell-Larcker criterion

	ATT	INT	PEOU	PR	PT	PU
ATT	0.846					
INT	0.763	0.914				
PEOU	0.577	0.549	0.831			
PR	-0.118	-0.172	0.070	0.830		
PT	0.635	0.719	0.502	-0.176	0.857	
PU	0.716	0.635	0.617	-0.016	0.540	0.863

Source: Author's computation

The HTMT criterion, proposed by Henseler et al. (2014), is considered a superior measure of discriminant validity because it offers greater specificity and sensitivity than other approaches. HTMT values close to 1 indicate the absence of discriminant validity, whereas studies have suggested thresholds of 0.90 (Gold et al., 2001). The HTMT values in Table 4.5 range from 0.042 to 0.862, which is below the 0.90 threshold. Thus, the discriminant validity is established.

Table 4.5: Heterotrait-Monotrait ratio (HTMT)

	ATT	INT	PEOU	PR	PT	PU
ATT						
INT	0.862					
PEOU	0.669	0.627				
PR	0.131	0.194	0.088			
PT	0.722	0.805	0.576	0.207		
PU	0.813	0.708	0.707	0.042	0.603	

Source: Author's computation

4.3 Structural Model Analysis

4.3.1 Multicollinearity

Multicollinearity was diagnosed using the Variance Inflation Factor (VIF), as shown in Table 4.6. Kim (2019) notes that multicollinearity occurs when the VIF exceeds 5. The inner-model VIFs for the factors range from 1.000 to 2.051, satisfying the threshold. Thus, the structural model is free from multicollinearity.

Table 4.6: Inner model variance inflation factor of the variables

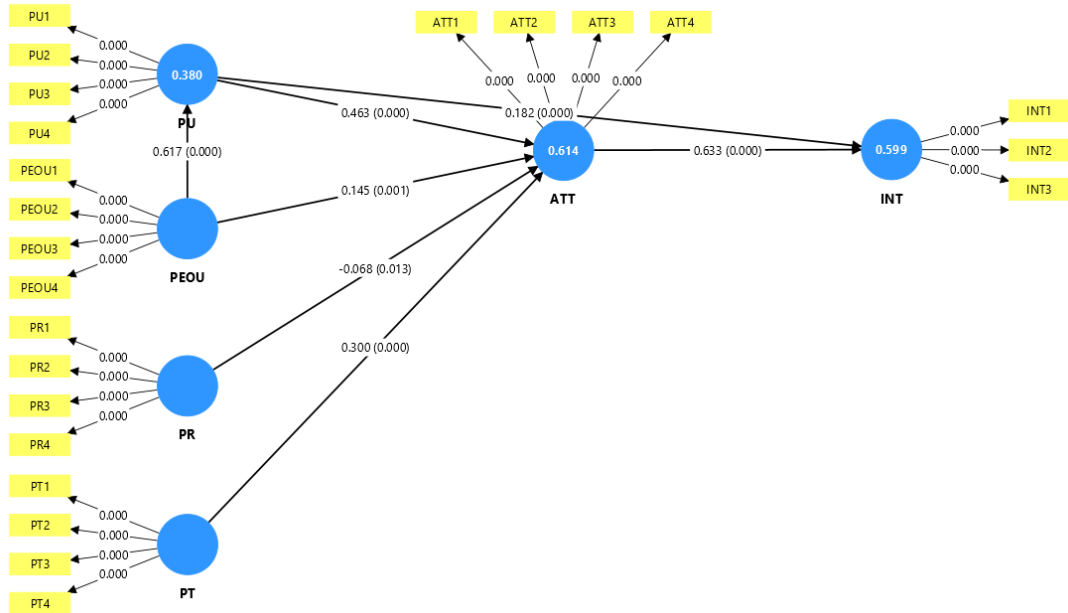
	VIF
ATT → INT	2.051
PEOU → ATT	1.772
PEOU → PU	1.000
PR → ATT	1.069
PT → ATT	1.597
PU → ATT	1.823
PU → INT	2.051

Source: Author's computation

4.3.2 Significance and Path Coefficients

This study uses the 5000-subsample bootstrapping function in SmartPLS 4, version 4.1.1.4, to examine the path relationships. The graphical output is displayed in Figure 4.7.

Figure 4.7: Structural model graphical output.



Source: Author's computation

The bootstrapping results in Table 4.8 indicate that Perceived Usefulness had a significant, positive effect on Intention to Adopt AI in banking ($\beta = 0.182$) and Attitude towards AI in banking ($\beta = 0.463$). Additionally, Perceived Ease of Use had a significant, positive effect on Perceived Usefulness ($\beta = 0.617$) and Attitude towards

AI in Banking ($\beta = 0.145$). Perceived Risk had a significant, negative effect on Attitude towards AI in Banking ($\beta = -0.068$), though the effect was mild. Perceived Trust had a significant and positive effect on Attitude towards AI in Banking ($\beta = 0.300$).

As for the mediation analysis, Attitude towards AI in Banking is a significant partial mediator between Perceived Usefulness and Intention to Adopt AI in Banking ($\beta = 0.293$). Conversely, Attitude is a significant mediator between Perceived Ease of Use and Intention ($\beta = 0.092$), Perceived Risk and Intention ($\beta = -0.043$), and Perceived Trust and Intention ($\beta = 0.190$). Additionally, Perceived Usefulness is a significant mediator between Perceived Ease of Use and Attitude towards AI in Banking ($\beta = 0.286$) and between Perceived Ease of Use and Intention ($\beta = 0.112$). Furthermore, Perceived Usefulness and Attitude towards AI in Banking jointly mediate the relationship between Perceived Ease of Use and Intention to Adopt AI in Banking ($\beta = 0.181$). Overall, the model is moderately fit, with an R-square of 0.599, explaining 59.9% of the variance in Intention to Adopt AI in banking.

Table 4.8: Structural model analysis

<i>Path</i>	<i>Coefficient</i>	<i>Std Dev</i>	<i>t-stats</i>	<i>p-values</i>	<i>Significance</i>
PU → INT	0.182	0.048	3.768	0.000	Significant
PU → ATT	0.463	0.042	10.923	0.000	Significant
PEOU → PU	0.617	0.030	20.589	0.000	Significant
PEOU → ATT	0.145	0.043	3.417	0.001	Significant
PR → ATT	-0.068	0.027	2.475	0.013	Significant
PT → ATT	0.300	0.035	8.517	0.000	Significant
<i>Mediation Analysis</i>					
PU → ATT → INT	0.293	0.033	8.925	0.000	Significant
PEOU → ATT → INT	0.092	0.029	3.188	0.001	Significant
PEOU → PU → ATT	0.286	0.029	9.897	0.000	Significant
PEOU → PU → INT	0.112	0.032	3.525	0.000	Significant
PEOU → PU → ATT → INT	0.181	0.021	8.751	0.000	Significant
PR → ATT → INT	-0.043	0.019	2.291	0.022	Significant
PT → ATT → INT	0.190	0.028	6.772	0.000	Significant

Source: Author's computation

4.3.3 Hypothesis Testing and Effect Size

The hypothesis-testing results summarised in Table 4.9 indicate that Perceived Usefulness significantly influences Intention to Adopt AI in Banking ($p = 0.000$). Therefore, H_1 is supported. Perceived Usefulness significantly influences Intention through Attitude towards AI in Banking ($p = 0.000$). Hence, H_2 is supported. Perceived Ease of Use influences Intention through Attitude ($p = 0.001$) and through Perceived Usefulness ($p = 0.000$). Thus, H_3 and H_4 are supported. Perceived Risk is influencing Intention through Attitude ($p = 0.022$). Therefore, H_5 is supported. Perceived Trust influences Intention through Attitude ($p = 0.000$). Thus, H_6 is supported.

Table 4.9: Hypothesis testing

	<i>Path</i>	<i>Coefficient</i>	<i>Std Dev</i>	<i>t-stats</i>	<i>p-values</i>	<i>Results</i>
H_1	PU $\rightarrow$ INT	0.182	0.048	3.768	0.000	Supported
H_2	PU $\rightarrow$ ATT $\rightarrow$ INT	0.293	0.033	8.925	0.000	Supported
H_3	PEOU $\rightarrow$ ATT $\rightarrow$ INT	0.092	0.029	3.188	0.001	Supported
H_4	PEOU $\rightarrow$ PU $\rightarrow$ INT	0.112	0.032	3.525	0.000	Supported
H_5	PR $\rightarrow$ ATT $\rightarrow$ INT	-0.043	0.019	2.291	0.022	Supported
H_6	PT $\rightarrow$ ATT $\rightarrow$ INT	0.190	0.028	6.772	0.000	Supported

Source: Author's computation

According to Cohen (1990) and Hair et al. (2014), the effect size can be categorised into large ($f^2 \geq 0.35$), medium ($f^2 \geq 0.15$), and small ($f^2 \geq 0.02$). The F-squares for each path are presented in Table 4.10. Two paths are found to have a large effect. They are Attitude towards AI in Banking to Intention to Adopt AI in Banking ($f^2 = 0.486$), and Perceived Ease of Use towards Perceived Usefulness ($f^2 = 0.614$). Conversely, Perceived Usefulness had a medium effect towards Attitude ($f^2 = 0.305$). Three paths showed small effects: Perceived Ease of Use towards Attitude ($f^2 = 0.031$), Perceived Trust towards Attitude ($f^2 = 0.146$), and Perceived Usefulness towards Intention ($f^2 = 0.04$). Notably, Perceived Risk has a minimal effect on Attitude ($f^2 = 0.011$).

Table 4.10: F-square of the path relationships

<i>Path</i>	<i>f-square</i>
ATT → INT	0.486
PEOU → ATT	0.031
PEOU → PU	0.614
PR → ATT	0.011
PT → ATT	0.146
PU → ATT	0.305
PU → INT	0.040

Source: Author's computation

The total effects of each variable towards Attitude towards AI in Banking and Intention to Adopt AI in Banking are summarised in Table 4.11. Perceived Ease of Use ($\beta = 0.431$), Perceived Risk ($\beta = -0.068$), Perceived Trust ($\beta = 0.300$), and Perceived Usefulness ($\beta = 0.463$) are significant factors influencing Attitude. Additionally, all five variables are statistically significant in influencing Intention, but their effect sizes vary. Attitude towards AI in Banking is identified as the predominant factor that influences Intention positively, with a coefficient of 0.633. Perceived Usefulness ranks second, with a coefficient of 0.475. The third is Perceived Ease of Use, with a coefficient of 0.385, followed by Perceived Trust, with a coefficient of 0.190. Perceived Risk is found to be the only negative factor with the minimal effect of -0.043 in the coefficient.

Table 4.11: Total effects of the variables towards ATT and INT

<i>Path</i>	<i>Total effects</i>	<i>p-values</i>
PEOU → ATT	0.431	0.000
PR → ATT	-0.068	0.013
PT → ATT	0.300	0.000
PU → ATT	0.463	0.000
ATT → INT	0.633	0.000
PEOU → INT	0.385	0.000
PR → INT	-0.043	0.022
PT → INT	0.190	0.000
PU → INT	0.475	0.000

Source: Author's computation

4.4 Chapter Summary

This study collected responses from 633 Gen Z Malaysians through convenience sampling during the sampling period. The construct is both reliable and valid. The multicollinearity diagnosis indicates that the model is free of multicollinearity. All path relationships in the model are significant, and all six hypotheses are supported. Nevertheless, the effect of Perceived Risk towards Attitude is minimal.

CHAPTER 5

DISCUSSION AND CONCLUSION

5.0 Introduction

This chapter discusses the implications of the results, outlines the limitations, and provides recommendations for future research. Sections 5.1 discuss the results consistent with the reviewed literature. Section 5.2 elaborates on the theoretical and practical implications. Section 5.3 examines the study's limitations, and Section 5.4 proposes suggestions to address the limitations.

5.1 Discussions of Results

The empirical findings suggest that all five factors in the model significantly influence Gen Z Malaysian's intention to adopt AI in banking. The model exhibits moderate explanatory power, explaining 59.9% of the variance in AI adoption intentions. The summary of the hypothesis-testing results is presented in Table 5.1.

Table 5.1: Summary of Hypothesis Testing Results

<i>Hypotheses</i>	<i>Results</i>
H ₁ : Perceived Usefulness (PU) positively influences Intention to Adopt AI in Banking (INT)	Supported
H ₂ : Perceived Usefulness (PU) positively influences Intention to Adopt AI in Banking (INT) through Attitude towards AI in Banking (ATT)	Supported
H ₃ : Perceived Ease of Use (PEOU) influences Intention to Adopt AI in Banking (INT) through Attitude towards AI in Banking (ATT)	Supported
H ₄ : Perceived Ease of Use (PEOU) influences Intention to Adopt AI in Banking (INT) through Perceived Usefulness (PU)	Supported
H ₅ : Perceived Risk (PR) influences Intention to Adopt AI in Banking (INT) through Attitude towards AI in Banking (ATT)	Supported
H ₆ : Perceived Trust (PT) influences Intention to Adopt AI in Banking (INT) through Attitude towards AI in Banking (ATT)	Supported

The PLS-SEM analysis results suggest that all hypothesised relationships are significant. The test result supports H₁, indicating that Perceived Usefulness positively influences Intentions to Adopt AI in Banking. This conclusion aligns with Afshan et al. (2018), Juwaheer et al. (2012), Noreen et al. (2023), Ryu (2017), and Singh et al.

(2024). It implies that the perceived benefits Gen Z Malaysians expect from adopting AI in banking are crucial in shaping their adoption behaviour. The benefits they expect include convenience, efficiency and effectiveness. Time-saving technology that meets their service needs appears to be the crucial factor that Gen Z Malaysians consider before adopting a new technology.

H₂ is supported, suggesting that Perceived Usefulness has a positive indirect influence on Intention to Adopt AI in Banking through Attitude towards AI in Banking. This finding is supported by Hu et al. (2019) and Belanche et al. (2019). It illustrates that Perceived Usefulness is an important element that shapes Gen Z Malaysian's attitude towards banking AI. Consequently, their attitude influences their likelihood to adopt banking AI. Therefore, efforts to shape a positive perception of banking AI among Gen Z Malaysians are essential to boosting its adoption in Malaysia.

Perceived Ease of Use is identified to positively influence Intention to Adopt AI in Banking through Attitude in the third hypothesis and through Perceived Usefulness in the fourth hypothesis. The results align with those of Belanche et al. (2019) and Sabir et al. (2023). The minimal effort required to become an expert in banking AI leads to a positive perception among consumers, who can easily benefit from its advantages. The positive perception is expressed as an affirmative attitude towards banking AI that stimulates them to adopt this technology. Additionally, the ease of familiarising oneself with banking AI functions reflects the immediate time cost required for future time savings. Hence, the easier it is to master banking AI, the more useful it becomes to Gen Z Malaysians. Consequently, the perceived usefulness becomes a push force for them to adopt banking AI. Thus, initiatives to develop a user-friendly user interface are critical to enhancing the adoption rate of banking AI among Gen Z Malaysians. Thus, H₃ and H₄ are supported.

The fifth hypothesis is supported in this study, which states that Perceived Risk influences Intention to adopt AI in Banking through Attitude. This result is identical to that of Alshari and Lokhande (2022). The risks associated with financial, privacy, and cybersecurity are shifting Gen Z Malaysians away from adopting banking AI. The riskier banking AI appeared to be in the mindset of Gen Z Malaysians, the more negative their attitude towards this technology. Their unfavourable attitudes, in turn,

hinder the adoption of banking AI. Consequently, the underlying risk must be appropriately managed to promote the adoption of banking AI among Gen Z Malaysians.

The sixth hypothesis suggests that Perceived Trust influences Intention to Adopt AI in Banking through Attitude. The outcome aligns with Hoang et al. (2021) and Alshari and Lokhande (2022). As Gen Z Malaysians perceive banking AI as more trustworthy, they are more optimistic about this technology. The optimistic sentiment then strengthens their willingness to adopt banking AI. Therefore, actions must be taken to promote trust and confidence among Gen Z Malaysians, thereby ensuring a high level of intention to adopt banking AI.

Based on the magnitude of the variables' effects, some factors should be prioritised when striving to enhance banking AI adoption rates. As discussed previously, the factors, ranked from most to least important, are: Attitude towards AI in Banking, Perceived Usefulness, Perceived Ease of Use, Perceived Trust, and Perceived Risk. Hence, policymakers and banking AI service providers should implement strategies to improve Gen Z Malaysians' attitudes towards banking AI and thereby enhance adoption rates. Among the factors that influence Attitude, Perceived Usefulness should be given priority, followed by Perceived Ease of Use, Perceived Trust, and, lastly, Perceived Risk, given their total effects on attitude.

5.2 Implications

5.2.1 Theoretical Implication

This study strengthens the TAM by reaffirming the significance of the fundamental path relationships in explaining the adoption of banking AI technology among Gen Z Malaysians. Perceived Usefulness demonstrates both direct and indirect influences on Intention, with Attitude as the mediator, consonant with TAM propositions. Likewise, Perceived Ease of Use follows the proposed theoretical pathway, which influences Intention through Perceived Usefulness and Attitude. All the observed effect directions align with expectations, reinforcing the validity of TAM in the banking technology

context. Therefore, this study has provided a robust theoretical foundation for future research on the application of TAM to AI financial services technology adoption.

Beyond TAM, this study also applies Prospect Theory to explain the asymmetric influence of perceived benefits and perceived risks on the intention to adopt banking AI. The descriptive results indicate that the average scores for the Perceived Risk items are 3.367, 3.667, 3.622, and 3.610, with a combined average of approximately 3.567. In contrast, the scores for Perceived Usefulness items are 3.581, 4.055, 4.112, and 3.877, averaging approximately 3.906. It demonstrates comparatively greater perceived benefits than perceived risks among Gen Z Malaysians. According to Prospect Theory's value function, individuals assign greater psychological weight to factors with higher perceived value, whether positive or negative. The higher Perceived Usefulness score is justified by the more substantial total effects on Intention compared to Perceived Risk. This finding supports the idea that the perceived benefits of adopting banking AI outweigh the perceived risks, offering empirical support for Prospect Theory in the Banking 4.0 context.

Besides risk-benefit assessments, trust is pivotal in the analysis. The empirical results highlight trust as a critical driver of adoption intention. The questionnaire items assessed respondents' perceptions of banking AI's trustworthiness, reliability, safety, and privacy, which are critical measures for evaluating cognitive trust (Kaasinen, 2005). However, because banking AI is still in its developmental stage in Malaysia, most Gen Z Malaysians tend to rely more on affective trust, forming judgments based on emotions rather than empirical evidence. The significant positive indirect relationship between Perceived Trust and Intention supports the theoretical argument by Bahmanziari et al. (2003) that a 'leap of faith' is often essential when deciding on adopting an emerging technology under imperfect information.

In short, this study tests TAM, incorporates Prospect Theory, and integrates trust-based explanations, providing a comprehensive theoretical model for understanding Malaysian Gen Z's willingness to embrace AI in the banking sector.

5.2.2 Practical Implications

This study provides valuable insights for the Malaysian government and banks to stimulate the transformation towards Banking 4.0. The empirical findings conclude that, in addition to TAM variables, risk and trust are also important in technology adoption decisions. Consequently, banking AI service providers should not only focus on promoting beneficial, convenient banking AI to customers, but also on promoting security and trust.

Gamification strategies are among the most effective tools in promoting the perceived usefulness, perceived ease of use, and ultimately adoption intention of AI among this youthful generation (Eklesia & Wijayanti, 2025). Shah (2025) found that 68% of Gen Z users use the Unified Payments Interface (UPI) for its ease of use and cashback perks. They value the benefit they can claim from their financial engagement, thereby strengthening their perception of financial services' usefulness. For instance, banking AI providers can develop a financial learning platform similar to Bank of America's "Better Money Habits" program, which educates users about personal financial management (Better Money Habits, 2025). The platform utilises banking AI to interact with Gen Z and to design personalised financial planning courses and investment education for them. They can earn points to exchange for rewards by interacting with the AI daily. Through this process, they can gain knowledge and other perks, which enhance their perception of the usefulness of banking AI. Additionally, the personalised content provided, accurately tailored to their preference, can build trust and further stimulate their intention to adopt banking AI.

Wigley and Kantaria (2023) report that 73% of Gen Z respondents cite customer experience as a pivotal factor in their preferred digital financial services. Inevitably, a streamlined process is crucial to the customer experience. Banking AI providers must simplify the user interface and provide step-by-step, head-start tutorials for new users to become familiar with the AI tools. Initiatives to implement a one-shot banking process are feasible by utilising AI to set up a chatbot that learns and understands text or voice input from users and performs the desired services on their behalf. For instance, Ryt Bank, a next-generation AI-powered digital bank in Malaysia led by YTL Digital Capital Sdn Bhd, has introduced the Ryt AI chatbot that can perform tasks

based on user text input. Users need to instruct the AI on what to do and then let it handle the rest, making the process convenient and efficient. The AI should also provide support and troubleshooting to assist the users when any errors occur. Thus, Gen Zs will feel relieved when using banking AI and be more willing to adopt it.

The interplay between risk and trust of banking AI must be handled systematically by both the government and the banking AI service providers. Constantino (2025) stated that 72% of Gen Z distrusts AI-based security and is sceptical towards digital security. However, he highlights that it is not necessarily that AI is bad at handling risk, but rather that Gen Z lacks confidence in the system. According to this study, although AI-based security is considered safer than traditional password systems, 61% of Gen Z still have more confidence in old-fashioned passwords than in AI-based security. Therefore, initiatives should openly reduce risk to ensure Gen Z is aware of the security enhancements and builds trust in AI-based security. Some feasible solutions include producing a transparent data-usage report, showcasing successful fraud-detection cases, and communicating the financial losses resulting from errors. On the other hand, the government should develop a standard framework for the banking AI security measures. For instance, the Malaysian government should mandate that AI banking service providers obtain a license from the National Cyber Security Agency (NACSA) to operate. In this sense, users will have greater confidence in service providers, as the government has guaranteed their commitment to safeguarding customers.

As cybersecurity is challenging to evaluate, the government serves as the key player in setting clear guidelines and ensuring that banking AI service providers comply with them. Beyond the Cybersecurity Act 2024, the Malaysian government should publish additional, stricter laws to govern the financial industry more effectively. For instance, the government should ensure that banking AI service providers adhere to ISO 27001 to protect the organisation's information assets (DNV, n.d.). Additionally, the government should involve users in the supervision framework by enhancing their cybersecurity literacy. Campaigns such as national literacy programmes should be conducted to equip Gen Zs with the essential cybersecurity knowledge needed to perform their monitoring role as users, ultimately reducing cybersecurity risks (Aldawood & Skinner, 2018). Simultaneously, as Gen Z gains sufficient cybersecurity

knowledge, they will be more objective in evaluating the risk and trustworthiness of banking AI. They are more likely to understand the risk-return trade-off of adopting banking AI, and as a result, they will be more willing to adopt it than stick with traditional methods.

Education for students is essential for cultivating talent that can contribute to a more mature cybersecurity ecosystem, not only in the financial industry but across the entire economy (Oroni et al., 2024). The government should provide student-focused digital finance education to nurture professionals who can help reduce exposure to banking AI risk and, thereby, promote trustworthiness. This educational programme should include both secondary and tertiary students to cultivate talents from an early stage. For example, the Financial Education Network (FEN) should expand its collaboration with Malaysia's Ministry of Education to establish a compulsory financial technology course in schools. Consequently, students' financial knowledge can be improved, thereby advancing progress toward achieving the vision of the National Strategy for Financial Literacy 2026-2030 (NS2.0) (Security Commission Malaysia, 2025). Besides, the government can host a national financial technology forum, similar to the Financial Technology Forum (FTF), and invite students to connect with industry professionals (Institutional Investor, n.d.). Consequently, they will have a foundational exposure towards the topic and can develop specialised expertise in this field in the future. The opportunity for students to participate in the Global Cybersecurity Forum (GCF) annual meeting and access their E-learning programs is also beneficial for younger generations, helping them grow and succeed (Global Cybersecurity Forum, n.d.). Thus, the government must formulate a pragmatic educational plan to ensure long-term success in reducing banking technology risk and increasing public trust.

5.3 Limitations

While this study offers valuable perspectives, it also has various constraints. Firstly, the scope of the study is limited to the adoption intention for banking AI among Gen Z Malaysians. Adoption intentions for different Banking 4.0 technologies may differ from those reported in this study. Additionally, behavioural intentions across generations or populations can vary significantly. Hence, this study may lack representativeness to explain adoption intention in other contexts.

Secondly, the convenience sampling method introduces sampling bias into this study. As questionnaire distribution depends on the researcher's own connections and social media targeting, there may be errors in reaching non-target respondents. For example, responses from non-Malaysians or non-Gen Z individuals may be included in the dataset despite clear eligibility criteria being outlined. Moreover, the respondents are predominantly tertiary students, which may limit the generalisability of the results. Such a phenomenon also raises concerns about heavy data clustering, where samples are collected only in a single area, such as the Klang Valley's universities. Thus, the flaws of non-random sampling are limiting the study's robustness.

Thirdly, banking AI technology is still in its developmental stage in Malaysia, with some pioneers adopting it and others not. Individuals who have used banking AI may have a different opinion from those without experience. The level of knowledge and awareness gained from active participation in this new technology influences their perception of banking AI. However, this study did not separate the samples by this characteristic because the questionnaire lacked items to capture such data. Findings from mixed samples may lead to inaccurate predictions of adoption intention.

Fourthly, as numerous factors influence individual decision-making, this study may overlook some significant variables. Although the model's goodness-of-fit is nearly 60%, it is not satisfactory. Some other factors proposed in the UTAUT 2 or other models may also influence adoption intention in our case. For example, multiple studies have cited the significance of subjective norms, motivational factors, technology performance, social factors, experience and facilitating conditions (Ajzen, 2011; Davis et al., 1992; Marikyan & Papagiannidis, 2025; Taylor & Todd, 1995; Thompson et al., 1991; Venkatesh et al., 2003). Therefore, there is a potential omitted variable bias.

Lastly, this research's cross-sectional nature limits the robustness of the findings over time. The rapid transformation towards Banking 4.0 and the development of banking AI in Malaysia are significantly shaping public perception. The empirical results based on 2025 data may be unreliable for inferring the population's perceptions in subsequent years. Hence, studies conducted at different time points are required to ensure timeliness and accuracy.

5.4 Suggestions for Future Research

Given the study's limited representativeness, future research should examine other contexts to validate and strengthen the model's reliability. For example, a similar study can be conducted among Gen Zs in other countries or a different age group in Malaysia. Additionally, various technologies, such as AI robo-advisors, neo-banks, peer-to-peer lending, and cryptocurrencies, can be assessed to enhance the robustness of this model.

Given the inherent sampling bias in non-random sampling, researchers should focus on refining their sampling methods. One recommended initiative is ensuring a balanced distribution of properties across the samples. For instance, if the male-to-female ratio of Gen Zs in Malaysia is 55:45, the study should obtain a sample with the same ratio. Such a proportion must also be applied to other traits, such as age bands, geographical regions, and educational levels. If population proportion data are unavailable, the researcher can employ quota sampling to maintain a well-diversified sample. For example, setting a quota of 200 males, 200 females, with 200 working Gen Zs, and 200 non-working Gen Zs. Future researchers should also include screening questions to filter out non-eligible responses, ensuring the data is free from invalid responses. Furthermore, researchers can use post-collection weighting to correct imbalanced responses by bringing the sample closer to the population weights.

As adopters and non-adopters may hold different opinions on adopting new technology, future researchers should strive to differentiate between them. The feasible solutions include setting an item in a questionnaire to identify adopters and non-adopters. Later, use a dummy variable to assess differences between the two groups. Future studies can also perform separate analyses for the two distinct groups to examine differences in the results.

Given the model's imperfect explanatory power, future studies are recommended to review more past literature to identify other potential variables that can be incorporated into the model. This includes studying other widely proposed models, such as UTAUT 2, the Theory of Planned Behaviour, the Diffusion of Innovation Theory, and Initial Trust Theory. The objective is to identify variables that are significant across multiple cases but are not included in this study. For instance, habit, subjective norm, trialability, compatibility, and competence.

Tackling the shortfall of cross-sectional studies can be done in several ways. Future researchers should employ a pre-test or post-test experimental design to validate causal relationships that have not been identified. Additionally, future researchers can conduct a longitudinal panel study to track behavioural changes over time, specifically examining risk and trust formation as banking AI matures. A multi-group analysis that categorises respondents by AI use frequency is a feasible strategy for simulating the time effect on behaviour. For example, when perceived trust is less influential among novices but more important among experts, one can expect that perceived trust will grow in importance as banking AI develop and more people become expert users.

5.5 Conclusion

As Malaysia undergoes rapid transformation towards Banking 4.0, understanding the public's perception of emerging financial technologies is essential to ensuring a smooth, successful transition. This study focuses on Banking AI, the key enabler of Banking 4.0, and on the readiness of Gen Z Malaysians, the most influential generational group shaping the nation's future landscape. Using primary data collected from 633 Gen Z respondents and analysing them using PLS-SEM, the study investigates the significant factors influencing Gen Z Malaysians' intention to adopt AI in banking.

By extending the TAM with risk and trust, the empirical findings suggest that all variables significantly influence Gen Z Malaysians' adoption intention. Perceived usefulness, perceived ease of use, perceived risk, and perceived trust shape Gen Z Malaysians' intention to adopt AI in banking through their attitude towards AI in banking. The findings support the notion that perceived risk has a relatively small total effect on Gen Z Malaysians' adoption intention. It implies that they are less concerned about potential risks when deciding whether to adopt AI banking services.

Although the results offer important insights for Malaysian banking AI service providers and the Malaysian government, the research has certain limitations. Issues regarding sampling randomness, model specification, and generalisability across time may deteriorate the robustness of the results. Therefore, additional research is crucial to validate these findings across contexts, including other Banking 4.0 technologies, diverse populations, and age groups.

REFERENCE

- Abdullah, D.B., Abdullah, M.Y., & Salleh, M.A.M. (2017). A review on the concept of Fourth Industrial Revolution and the government's initiatives to promote it among youths in Malaysia. *Journal of Social Sciences and Humanities*, 2(1), 1-8.
- Afshan, S., Sharif, A., Waseem, N., & Frooghi, R. (2018). Internet banking in Pakistan: an extended technology acceptance perspective. *International Journal of Business Information Systems*, 27(3), 383. <https://doi.org/10.1504/ijbis.2018.089863>
- Aggarwal, M., Nayak, K. M., & Bhatt, V. (2023). Examining the factors influencing fintech adoption behaviour of gen Y in India. *Cogent Economics & Finance*, 11(1). <https://doi.org/10.1080/23322039.2023.2197699>
- Ajzen, I. (2011). The theory of planned behaviour: Reactions and reflections. *Psychology & Health*. https://www.researchgate.net/profile/Icek_Ajzen/publication/51653805_The_theory_of_planned_behaviour_reactions_and_reflections/links/0f31753a05cf5d4de0000000.pdf
- Akrout, H., Diallo, M. F., Akrou, W., & Chandon, J. L. (2016). Affective trust in buyer-seller relationships: A two-dimensional scale. *Journal of Business & Industrial Marketing*, 31(2), 260–273. <https://doi.org/10.1108/JBIM-11-2014-0223>

- Alalwan, A. A., Dwivedi, Y. K., & Rana, N. P. (2017). Factors influencing adoption of mobile banking by Jordanian bank customers: Extending UTAUT2 with trust. *International Journal of Information Management*, 37(3), 99–110. <https://doi.org/10.1016/j.ijinfomgt.2017.01.002>
- Aldawood, H., & Skinner, G. (2018). Educating and raising awareness on cyber security social engineering: A literature review. *Conference: IEEE TALE Conference 2018*, 62–68. <https://doi.org/10.1109/tale.2018.8615162>
- Alejandro, C. C., & Güemes-Castorena, D. (2024). Trust and distrust coexistence and change in the adoption of an emerging technology in organizations. *Computers in Human Behavior Reports*, 16, 100503. <https://doi.org/10.1016/j.chbr.2024.100503>
- Alshammari, S. H., & Rosli, M. S. (2020, December 14). *A review of technology acceptance models and theories*. <http://161.139.21.34/itlj/index.php/itlj/article/view/56>
- Alshari, H. A., & Lokhande, M. A. (2022). The impact of demographic factors of clients' attitudes and their intentions to use FinTech services on the banking sector in the least developed countries. *Cogent Business & Management*, 9(1). <https://doi.org/10.1080/23311975.2022.2114305>
- Ashcraft, L. (2023, December 11). *Privacy concerns don't get in the way of Gen Z's enthusiasm about AI-enabled banking*. Emarketer. <https://www.emarketer.com/content/privacy-concerns-dont-get-in-way-of-gen-z-enthusiasm-about-ai-enabled-banking>

- Bagozzi, R. P., & Yi, Y. (1990). Assessing method variance in multitrait-multimethod matrices: The case of self-reported affect and perceptions at work. *Journal of Applied Psychology*, 75(5), 547–560. <https://doi.org/10.1037/0021-9010.75.5.547>
- Bahmanziari, T., Pearson, J. M., & Crosby, L. (2003). Is trust important in technology adoption? A policy capturing approach. *Journal of Computer Information Systems*, 43(4), 46–54. <https://doi.org/10.1080/08874417.2003.11647533>
- Bank Negara Malaysia (2022, April 29). *Five successful applicants for the digital bank licences*. Bank Negara Malaysia. Retrieved 30/4/2025 from <https://www.bnm.gov.my/-/digital-bank-5-licences>.
- Bank Negara Malaysia (2025, August). Artificial Intelligence in the Malaysian Financial Sector. Bank Negara Malaysia. <https://www.bnm.gov.my/-/dp-aifs25>
- Barberis, N. C. (2013). Thirty years of prospect theory in economics: A review and assessment. *The Journal of Economic Perspectives*, 27(1), 173–196. <https://doi.org/10.1257/jep.27.1.173>
- Bauer, R. A. (1967). Consumer behavior as risk taking. *Marketing: Critical perspectives on business and management*, 593, 13-21.
- Belanche, D., Casaló, L. V., & Flavián, C. (2019). Artificial Intelligence in FinTech: understanding robo-advisors adoption among customers. *Industrial Management & Data Systems*, 119(7), 1411–1430. <https://doi.org/10.1108/imds-08-2018-0368>

Better Money Habits (2025, June 23). Personal financial education from Better Money Habits. Better Money Habits.

<https://bettermoneyhabits.bankofamerica.com/en>

Bhandari, P. (2020). What is qualitative research? | Methods & examples. Scribbr.

<https://www.scribbr.com/methodology/qualitative-research/>

Bhatia, M. (2022). *Banking 4.0*. <https://doi.org/10.1007/978-981-16-6069-6>

Bhatt, A. & Bhatt, S. (2016). Factors affecting customer's adoption of mobile banking services. *Journal of Internet Banking and Commerce*, 21(1).

https://www.researchgate.net/publication/301680182_Factors_affecting_customer's_adoption_of_mobile_banking_services

Bryant, M. J., Wilde, S. J., Coles, G., Sorwar, G., & Bui, V. (2025). Mobile app usage and adoption through the lens of cognitive, affective and behavioral trust constructs. *International Journal of Human-Computer Interaction*, 1–25.

<https://doi.org/10.1080/10447318.2025.2531289>

Bryman, A. (2016). *Social research methods*. Oxford university press.

Chandra, S., Srivastava, S. C., & Theng, Y.-L. (2010). Evaluating the role of trust in consumer adoption of mobile payment systems: An empirical analysis.

Communications of the Association for Information Systems, 27, 561–588.

<https://doi.org/10.17705/1CAIS.02729>

Chin, W. W., Cheah, J.-H., Liu, Y., Ting, H., Lim, X.-J., & Cham, T. H. (2020).

Demystifying the role of causal-predictive modeling using partial least squares

- structural equation modeling in information systems research. *Industrial Management & Data Systems*, 120(12), 2161–2209.
- Chowdhury, S. (2005). The role of affect-and cognition-based trust in complex knowledge sharing. *Journal of Managerial Issues*, 17(3), 310–326.
<https://www.jstor.org/stable/40604504>
- Cohen, J. (1990). Statistical power analysis for the behavioral sciences. *Computers Environment and Urban Systems*, 14(1), 71. [https://doi.org/10.1016/0198-9715\(90\)90050-4](https://doi.org/10.1016/0198-9715(90)90050-4)
- Constantino, T. (2025, Apr 1). 72% of Gen Z distrust AI security—More than any other surveyed group. Forbes.
<https://www.forbes.com/sites/torconstantino/2025/04/01/72-of-gen-z-distrust-ai-security-more-than-any-other-surveyed-group/>
- Cummings, L. L., & Bromiley, P. (1996). The Organizational Trust Inventory (OTI): Development and validation. In *SAGE Publications, Inc. eBooks* (pp. 302–330).
<https://doi.org/10.4135/9781452243610.n15>
- Currall, S. C., & Judge, T. A. (1995). Measuring trust between organizational boundary role persons. *Organizational Behavior and Human Decision Processes*, 64(2), 151–170. <https://doi.org/10.1006/obhd.1995.1097>
- Da Silva, R. (2021). Calls for behavioural biometrics as bank fraud soars. *Biometric Technology Today*, 2021(9), 7–9. [https://doi.org/10.1016/s0969-4765\(21\)00095-3](https://doi.org/10.1016/s0969-4765(21)00095-3)

- Daqar, M. a. M. A., Arqawi, S., & Karsh, S. A. (2020). Fintech in the eyes of Millennials and Generation Z (the financial behavior and Fintech perception). *Banks and Bank Systems*, 15(3), 20–28. [https://doi.org/10.21511/bbs.15\(3\).2020.03](https://doi.org/10.21511/bbs.15(3).2020.03)
- Davis, F. D. (1985). A technology acceptance model for empirically testing new end-user information systems: Theory and results. <http://ci.nii.ac.jp/naid/20001062454>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319. <https://doi.org/10.2307/249008>
- Davis, F. D., & Venkatesh, V. (1996). A critical assessment of potential measurement biases in the technology acceptance model: three experiments. *International Journal of Human-Computer Studies*, 45(1), 19–45. <https://doi.org/10.1006/ijhc.1996.0040>
- Davis, F. D., & Venkatesh, V. (2004). Toward preprototype user acceptance testing of new information systems: Implications for software project management. *IEEE Transactions on Engineering Management*, 51(1), 31–46. <https://doi.org/10.1109/tem.2003.822468>
- Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1992). Extrinsic and intrinsic motivation to use computers in the workplace1. *Journal of Applied Social Psychology*, 22(14), 1111–1132. <https://doi.org/10.1111/j.1559-1816.1992.tb00945.x>

DNV (n.d.). *Why ISO 27001 certification matters and how training drives success.*

Det Norske Veritas. <https://www.dnv.com/ca/articles/20250120-why-iso-27001-certification-matters-and-how-training-drives-success/>

Egilsdottir, T. (2023, February 15). *Gen Z and how they bank?*

<https://www.spuerkeess.lu/en/blog/experts-corner/gen-z-and-how-they-bank/>

Eklesia, J. & Wijayanti, C. A. (2025). The impact of gamification on gen z's behavioral intention through e-commerce shopee: Does reward moderate the relationship?

The 4th International Conference on Entrepreneurship.

https://www.researchgate.net/publication/389045792_The_Impact_of_Gamification_on_Gen_Z's_Behavioral_Intention_through_E-Commerce_Shopee_Does_Reward_Moderate_the_Relationship

Elfil, M., & Negida, A. (2017). Sampling methods in Clinical Research; an Educational Review. *Emergency (Tehran, Iran)*, 5(1), e52.

Fasnacht, D. (2021). Banking 4.0: Digital ecosystems and Super-Apps. In *Sustainable finance* (pp. 235–256). https://doi.org/10.1007/978-3-030-52275-9_15

Featherman, M. S., & Pavlou, P. A. (2003). Predicting e-services adoption: A perceived risk facets perspective. *International Journal of Human-Computer Studies*, 59(4), 451–474. [https://doi.org/10.1016/s1071-5819\(03\)00111-3](https://doi.org/10.1016/s1071-5819(03)00111-3)

Financial Education Network (n.d.). Strategi literasi kewangan kebangsaan Malaysia 2026-2030. Financial Education Network. <https://www.fenetwork.my/>

- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39. <https://doi.org/10.2307/3151312>
- Francis, J. C. (2024). Harry Markowitz's contributions to utility theory. *Annals of Operations Research*, 346(1), 113–125. <https://doi.org/10.1007/s10479-024-06210-2>
- Garg, P., Gupta, B., Chauhan, A. K., Sivarajah, U., Gupta, S., & Modgil, S. (2020). Measuring the perceived benefits of implementing blockchain technology in the banking sector. *Technological Forecasting and Social Change*, 163, 120407. <https://doi.org/10.1016/j.techfore.2020.120407>
- Ghanad, A. (2023). An overview of quantitative research methods. *International Journal of Multidisciplinary Research and Analysis*, 06(08). <https://doi.org/10.47191/ijmra/v6-i8-52>
- Global Cybersecurity Forum (n.d.). GCF annual meeting 2025. Global Cybersecurity Forum. <https://gcforum.org/en/events/423f93a5-35bf-4043-ab23-2d39cf5de50e/>
- Gold, A. H., Malhotra, A., & Segars, A. H. (2001). Knowledge management: An organizational capabilities perspective. *Journal of Management Information Systems*, 18(1), 185–214. <https://doi.org/10.1080/07421222.2001.11045669>
- Government of Malaysia. (n.d). *Population Table: Malaysia* | data.gov.my. [data.gov.my](https://data.gov.my/data-catalogue/population_malaysia). https://data.gov.my/data-catalogue/population_malaysia

- Grabner-Kräuter, S., & Faullant, R. (2008). Consumer acceptance of internet banking: the influence of internet trust. *International Journal of Bank Marketing*, 26(7), 483–504. <https://doi.org/10.1108/02652320810913855>
- Hair, J. F., Hollingsworth, C. L., Randolph, A. B., & Chong, A. Y. L. (2017). An updated and expanded assessment of PLS-SEM in information systems research. *Industrial Management & Data Systems*, 117(3), 442–458.
- Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2014). A primer on partial least squares structural equation modeling (PLS-SEM). <http://ci.nii.ac.jp/ncid/BB15179462>
- Hair, J. F., Hult, G. T. M., Ringle, C. M., Sarstedt, M., Danks, N. P., & Ray, S. (2021). An introduction to structural equation modeling. In *Classroom companion: business* (pp. 1–29). https://doi.org/10.1007/978-3-030-80519-7_1
- Hair, J. F., Sarstedt, M., & Ringle, C. M. (2019). Rethinking some of the rethinking of partial least squares. *European Journal of Marketing*, 53(4), 566–584.
- Hair, J., & Alamer, A. (2022). Partial Least Squares Structural Equation Modeling (PLS-SEM) in second language and education research: Guidelines using an applied example. *Research Methods in Applied Linguistics*, 1(3), 100027. <https://doi.org/10.1016/j.rmal.2022.100027>
- Hassan, H. E., & Wood, V. R. (2019). Does country culture influence consumers' perceptions toward mobile banking? A comparison between Egypt and the United States. *Telematics and Informatics*, 46, 101312. <https://doi.org/10.1016/j.tele.2019.101312>

- Hassan, Z. A., Schattner, P., & Mazza, D. (2006). Doing a pilot study: Why is it essential?. *Malays Fam Physician*, 1(2-3), 70–73.
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2014). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135. <https://doi.org/10.1007/s11747-014-0403-8>
- Hisham, H. (2024, January 26). *The transformational Impact of Generative AI on FinTech Companies*. JurisTech. <https://juristech.net/juristech/transformational-impact-generative-ai-fintech/>
- Hoang, Y. H., Nguyen, D. T., Tran, L. H., Nguyen, N. T., & Vu, N. B. (2021). Customers' adoption of financial services offered by banks and fintechs partnerships: evidence of a transitional economy. *Data Science in Finance and Economics*, 1(1), 77–95. <https://doi.org/10.3934/dsfe.2021005>
- Hoo, W. C., Marthyna, A., Wolor, C. W., Tan, G. P. L., & Prompanyo, M. (2024). Factors influencing intention to accept digital banks among generation Z in Malaysia. *Journal of Lifestyle and SDGs Review*, 4(3), e01815. <https://doi.org/10.47172/2965-730x.sdgsreview.v4.n03.pe01815>
- Hu, Z., Ding, S., Li, S., Chen, L., & Yang, S. (2019). Adoption intention of Fintech services for bank users: An empirical examination with an extended Technology Acceptance Model. *Symmetry*, 11(3), 340. <https://doi.org/10.3390/sym11030340>

- Huh, H. J., Kim, T., & Law, R. (2008). A comparison of competing theoretical models for understanding acceptance behavior of information systems in upscale hotels. *International Journal of Hospitality Management*, 28(1), 121–134. <https://doi.org/10.1016/j.ijhm.2008.06.004>
- Hurani, J., & Abdel-Haq, M. K. (2025). Factors influencing FinTech adoption among bank customers in Palestine: An extended Technology Acceptance model approach. *International Journal of Financial Studies*, 13(1), 11. <https://doi.org/10.3390/ijfs13010011>
- Ikhsan, R. B., Fernando, Y., Prabowo, H., Yuniarty, N., Gui, A., & Kuncoro, E. A. (2024). An empirical study on the use of artificial intelligence in the banking sector of Indonesia by extending the TAM model and the moderating effect of perceived trust. *Digital Business*, 100103. <https://doi.org/10.1016/j.digbus.2024.100103>
- Institutional Investors (n.d.). *Financial technology forum: the premier network for technology leaders from leading investment management firms*. <https://www.institutionalinvestor.com/Financial-Technology-Forum>
- Johnson, D., & Grayson, K. (2005). Cognitive and affective trust in service relationships. *Journal of Business Research*, 58(4), 500–507. [https://doi.org/10.1016/S0148-2963\(03\)00140-1](https://doi.org/10.1016/S0148-2963(03)00140-1)
- Juwaheer, T. D., Pudaruth, S., & Ramdin, P. (2012). Factors influencing the adoption of internet banking: A case study of commercial banks in Mauritius. *World Journal of Science Technology and Sustainable Development*, 9(3), 204–234. <https://doi.org/10.1108/20425941211250552>

- Kaasinen, E. (2005). User acceptance of mobile services: Value, ease of use, trust and ease of adoption. VTT Technical Research Centre of Finland Espoo. <https://publications.vtt.fi/pdf/publications/2005/P566.pdf>
- Kahneman, D. (2011). *Thinking, fast and slow*. <http://ci.nii.ac.jp/ncid/BB2184891X>
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263. <https://doi.org/10.2307/1914185>
- Kaya, O., Schildbach, J., AG, D. B., & Schneider, S. (2019). Artificial intelligence in banking. *Artificial intelligence*.
- Khan, W. U., Shachak, A., & Seto, E. (2022). Understanding decision-making in the adoption of digital health technology: The role of behavioral economics' prospect theory. *Journal of Medical Internet Research*, 24(2), e32714. <https://doi.org/10.2196/32714>
- Kim, D., & Bae, J. K. (2020). Understanding internet-only bank service adoption: An integration of the Unified Technology Theory of Acceptance and Innovation Resistance Model. *Global Business & Finance Review*, 25(3), 49–59. <https://doi.org/10.17549/gbfr.2020.25.3.49>
- Kim, G., Shin, B., & Lee, H. G. (2009). Understanding dynamics between initial trust and usage intentions of mobile banking. *Information Systems Journal*, 19(3), 283–311. <https://doi.org/10.1111/J.1365-2575.2007.00269.X>
- Kim, J. H. (2019). Multicollinearity and misleading statistical results. *Korean Journal of Anesthesiology*, 72(6), 558–569. <https://doi.org/10.4097/kja.19087>

- Kim, K. J., & Sundar, S. S. (2016). Mobile persuasion: Can screen size and presentation mode make a difference to trust? *Human Communication Research*, 42(1), 45–70. <https://doi.org/10.1111/hcre.12064>
- Kok, S. L., & Siripipatthanakul, S. (2023, November 26). *Artificial intelligence (AI) adoption: the case of the Malaysian financial industry*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4644825
- Komiak, S. Y., & Benbasat, I. (2006). The effects of personalization and familiarity on trust and adoption of recommendation agents. *MIS Quarterly*, 30(4), 941–960. <https://doi.org/10.2307/25148760>
- KPMG (2020). *Digital banking: The inclusive agenda*. KPMG. <https://assets.kpmg.com/content/dam/kpmg/my/pdf/digitalbanking-report.pdf>
- KPMG (2024). *Customer experience excellence report 2023-24*. KPMG. <https://assets.kpmg.com/content/dam/kpmg/my/pdf/cee-malaysia-report-2023-24.pdf>
- Krejcie, R. V., & Morgan, D. W. (1970). Determining sample size for research activities. *Educational and Psychological Measurement*, 30(3), 607–610. <https://doi.org/10.1177/001316447003000308>
- Kumar, S., & Dami, M. (2021). Integrating Diffusion of innovation to Technology Acceptance Model: A survey of millennials' intention to use E-Money Card. *Advances in Economics, Business and Management Research/Advances in Economics, Business and Management Research*, 177. <https://doi.org/10.2991/aebmr.k.210522.026>

- Lancaster, G. A., Dodd, S., & Williamson, P. R. (2004). Design and analysis of pilot studies: recommendations for good practice. *Journal of Evaluation in Clinical Practice*, 10(2), 307–312. <https://doi.org/10.1111/j..2002.384.doc.x>
- Latan, H., & Noonan, R. (2017). *Partial least squares path modeling*. <https://doi.org/10.1007/978-3-319-64069-3>
- Lee, J., & Chen, X. (2022). Exploring users' adoption intentions in the evolution of artificial intelligence mobile banking applications: The intelligent and anthropomorphic perspectives. *International Journal of Bank Marketing*, 40(4), 631–658. <https://doi.org/10.1108/ijbm-08-2021-0394>
- Lee, Y., Kozar, K. A., & Larsen, K. R. (2003). The Technology Acceptance Model: Past, present, and future. *Communications of the Association for Information Systems*, 12. <https://doi.org/10.17705/1cais.01250>
- Levy, J. S. (1992). An introduction to Prospect Theory. *Political Psychology*, 13(2), 171–186. <http://www.jstor.org/stable/3791677>
- Lewis, J. D., & Weigert, A. (1985). Trust as a social reality. *Social Forces*, 63(4), 967–985. <https://doi.org/10.1093/sf/63.4.967>
- Lewis, J. D., & Weigert, A. J. (2012). The social dynamics of trust: Theoretical and empirical research, 1985–2012. *Social Forces*, 91(1), 25–31. <https://doi.org/10.1093/sf/sos116>
- Li, H., Wu, J., Gao, Y., & Shi, Y. (2016). Examining individuals' adoption of healthcare wearable devices: An empirical study from privacy calculus perspective.

International Journal of Medical Informatics, 88, 8–17.
<https://doi.org/10.1016/j.ijmedinf.2015.12.010>

Li, M., Lin, S., Chan, Y., & Wu, C. (2021). Customer involvement facets stimulating customers' intention to use Internet-Only Bank services in China. *Journal of Organizational and End User Computing*, 33(5), 74–97.
<https://doi.org/10.4018/joeuc.20210901.oa4>

Lockett, A., & Littler, D. (1997). The adoption of direct banking services. *Journal of Marketing Management*, 13(8), 791–811.
<https://doi.org/10.1080/0267257x.1997.9964512>

Maja, M. M., & Letaba, P. (2022). Towards a data-driven technology roadmap for the bank of the future: Exploring big data analytics to support technology roadmapping. *Social Sciences & Humanities Open*, 6(1), 100270.
<https://doi.org/10.1016/j.ssaho.2022.100270>

Malhotra, N., Hall, J., Shaw, M., & Oppenheim, P. (2007). *Essentials of Marketing Research, An Applied Orientation*. P. Ed Custom Books, 2007. ISBN: 1486010792

Marangunić, N., & Granić, A. (2014). Technology acceptance model: a literature review from 1986 to 2013. *Universal Access in the Information Society*, 14(1), 81–95. <https://doi.org/10.1007/s10209-014-0348-1>

Marikyan, D. & Papagiannidis, S. (2025). *Unified Theory of Acceptance and Use of Technology: A review*. TheoryHub Book.

<https://open.ncl.ac.uk/theories/2/unified-theory-of-acceptance-and-use-of-technology/>

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, 20(3), 709–734.
<https://doi.org/10.5465/amr.1995.9508080335>

McKinsey & Company (2022, August 17). *What are Industry 4.0, the Fourth Industrial Revolution, and 4IR?*. McKinsey & Company.
<https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-are-industry-4-0-the-fourth-industrial-revolution-and-4ir#/>

McKinsey & Company (2024, August 28). *What is Gen Z?*. McKinsey & Company.
<https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-gen-z>

McKnight, D. H., & Chervany, N. L. (1996). The meanings of trust. Carlson School of Management, University of Minnesota.

McKnight, D. H., Choudhury, V., & Kacmar, C. (2002). Developing and validating trust measures for e-commerce: An integrative typology. *Information Systems Research*, 13(3), 334–359. <https://doi.org/10.1287/isre.13.3.334.81>

Mehdiabadi, A., Tabatabeinasab, M., Spulbar, C., Yazdi, A. K., & Birau, R. (2020). Are we ready for the challenge of Banks 4.0? Designing a roadmap for banking systems in Industry 4.0. *International Journal of Financial Studies*, 8(2), 32.
<https://doi.org/10.3390/ijfs8020032>

- Menezes, A. D., K, K., & Naik, S. G. (2024). Customer perception of artificial intelligence in public banking: An empirical analysis. *ITM Web of Conferences*, 68, 01026. <https://doi.org/10.1051/itmconf/20246801026>
- MIDA (2022, December 1). *How AI can power economic recovery and overcome challenges in Malaysia*. Malaysian Investment Development Authority. <https://www.mida.gov.my/mida-news/how-ai-can-power-economic-recovery-and-overcome-challenges-in-malaysia/>
- Moden, N., Coleman, S., & Savage, K. (2023, Jul 10). How banking on Gen Z talent will make or break the future of banking. EY. https://www.ey.com/en_gl/insights/banking-capital-markets/how-banking-on-gen-z-talent-will-make-or-break-the-future-of-banking
- Moorman, C., Zaltman, G., & Deshpande, R. (1992). Relationships between providers and users of market research: The dynamics of trust within and between organizations. *Journal of Marketing Research*, 29(3), 314. <https://doi.org/10.2307/3172742>
- Narang, A., Vashisht, P., & Bajaj, S. B. (2024). Artificial intelligence in banking and finance. *International Journal of Innovative Research in Computer Science & Technology*, 12(2), 130–134. <https://doi.org/10.55524/ijirest.2024.12.2.23>
- Nawaysch, M. K. A. (2020). FinTech in COVID-19 and beyond: What factors are affecting customers' choice of FinTech applications? *Journal of Open Innovation Technology Market and Complexity*, 6(4), 153. <https://doi.org/10.3390/joitmc6040153>

- Nel, J., & Boshoff, C. (2020). "I just don't like digital-only banks, and you should not use them either": Traditional-bank customers' opposition to using digital-only banks. *Journal of Retailing and Consumer Services*, 59, 102368. <https://doi.org/10.1016/j.jretconser.2020.102368>
- Nel, J., & Boshoff, C. (2021). Traditional-bank customers' digital-only bank resistance: evidence from South Africa. *International Journal of Bank Marketing*, 39(3), 429–454. <https://doi.org/10.1108/ijbm-07-2020-0380>
- Noor, N. R. A. M., & Mansor, N. (2019). Exploring the adaptation of artificial intelligence in whistleblowing practice of the internal auditors in Malaysia. *Procedia Computer Science*, 163, 434–439. <https://doi.org/10.1016/j.procs.2019.12.126>
- Nooteboom, B., & Six, F. (2003). *The trust process in organizations: Empirical studies of the determinants and the process of trust development*. Edward Elgar Publishing. <https://doi.org/10.4337/9781843767350>
- Noreen, U., Shafique, A., Ahmed, Z., & Ashfaq, M. (2023). Banking 4.0: Artificial Intelligence (AI) in banking industry & consumer's perspective. *Sustainability*, 15(4), 3682. <https://doi.org/10.3390/su15043682>
- Oney, E., Guven, G. O., & Rizvi, W. H. (2017). The determinants of electronic payment systems usage from consumers' perspective. *Economic Research-Ekonomska Istraživanja*, 30(1), 394–415. <https://doi.org/10.1080/1331677x.2017.1305791>

- Oroni, C. Z., Xianping, F., Ndunguru, D. D., & Ani, A. (2024). Enhancing cyber safety in e-learning environment through cybersecurity awareness and information security compliance: PLS-SEM and FsQCA analysis. *Computers & Security*, 150, 104276. <https://doi.org/10.1016/j.cose.2024.104276>
- Othman et al. (2019). Malaysian consumer behaviour towards Internet banking: An application of Technology Acceptance Model. *International Journal of Psychosocial Rehabilitation* 23(2):689-703. DOI:10.37200/IJPR/V23I2/PR190324
- Park, E. S., & Park, M. S. (2020). Factors of the Technology Acceptance Model for construction IT. *Applied Sciences*, 10(22), 8299. <https://doi.org/10.3390/app10228299>
- Patel, K. J., & Patel, H. J. (2017). Adoption of internet banking services in Gujarat. *International Journal of Bank Marketing*, 36(1), 147–169. <https://doi.org/10.1108/ijbm-08-2016-0104>
- Payne, E. M., Peltier, J. W., & Barger, V. A. (2018). Mobile banking and AI-enabled mobile banking. *Journal of Research in Interactive Marketing*, 12(3), 328–346. <https://doi.org/10.1108/jrim-07-2018-0087>
- Polit, D. F., & Beck, C. T. (2017). *Nursing research: Generating and assessing evidence for nursing practice (10th ed.)*. Philadelphia, PA: Wolters Kluwer/Lippincott Williams & Wilkins.

Propnex. (n.d.). *What's driving Gen Z & millennials in Malaysia's property market?*

<https://www.propnex.com.my/post/details/blog/gen%20z%20millenials%20real%20estate>

PwC. (n.d.). *Financial services horizons - looking ahead.* PwC.

<https://www.pwc.com/m1/en/industries/financial-services/financial-services-horizons-looking-ahead.html>

Pymnts. (2023, December 1). *Gen Z interested in AI-Enabled banking experiences*

despite privacy concerns. PYMNTS.com.

<https://www.pymnts.com/news/artificial-intelligence/2023/generation-z-interested-ai-enabled-banking-experiences-despite-privacy-concerns/>

Rahman, M., Ming, T. H., Baigh, T. A., & Sarker, M. (2021). Adoption of artificial intelligence in banking services: An empirical analysis. *International Journal of Emerging Markets*, 18(10), 4270–4300. <https://doi.org/10.1108/ijoem-06-2020-0724>

Rempel, J. K., Holmes, J. G., & Zanna, M. P. (1985). Trust in close relationships.

Journal of Personality and Social Psychology, 49(1), 95–112.

<https://doi.org/10.1037/0022-3514.49.1.95>

Roseline, J. F., Naidu, G., Pandi, V. S., Rajasree, S. A. A., & Mageswari, D. (2022).

Autonomous credit card fraud detection using a machine learning approach.

Computers & Electrical Engineering, 102, 108132.

<https://doi.org/10.1016/j.compeleceng.2022.108132>

- Rosli, M. S., Saleh, N. S., Ali, A. M., & Bakar, S. A. (2023). Factors determining the acceptance of e-wallet among gen z from the lens of the extended Technology Acceptance Model. *Sustainability*, 15(7), 5752. <https://doi.org/10.3390/su15075752>
- Ryu, H. (2018). What makes users willing or hesitant to use Fintech?: the moderating effect of user type. *Industrial Management & Data Systems*, 118(3), 541–569. <https://doi.org/10.1108/imds-07-2017-0325>
- Sabir, A. A., Ahmad, I., Ahmad, H., Rafiq, M., Khan, M. A., & Noreen, N. (2023). Consumer acceptance and adoption of AI robo-advisors in fintech industry. *Mathematics*, 11(6), 1311. <https://doi.org/10.3390/math11061311>
- Said, N. M., Zainuddin, S. A., Nasir, N. a. M., Abdullah, T., Yusoff, M. N. H., Yasoa, M. R., & Muhamad, S. F. (2022). The factor drive consumer perceptions towards banking 4.0. In *Lecture notes in networks and systems* (pp. 61–72). https://doi.org/10.1007/978-3-031-08087-6_5
- Saif, M. A. M., Hussin, N., Husin, M. M., Alwadain, A., & Chakraborty, A. (2022). Determinants of the intention to adopt Digital-Only banks in Malaysia: The extension of environmental concern. *Sustainability*, 14(17), 11043. <https://doi.org/10.3390/su141711043>
- Saif, M. A., Hussin, N., Husin, M. M., Muneer, A., & Alwadain, A. (2024). Beyond conventions: Unravelling perceived value's role in shaping digital-only banks' adoption. *Technological Forecasting and Social Change*, 203, 123337. <https://doi.org/10.1016/j.techfore.2024.123337>

- Salmasi, S. D., Sedighi, M., Sharif, H., & Shah, M. H. (2024). Adoption of new banking models from a consumer perspective: the case of Iran. *International Journal of Bank Marketing*, 42(7), 1946–1977. <https://doi.org/10.1108/ijbm-02-2023-0094>
- Sarmiento, R. P., & Costa, V. (2019). Confirmatory Factor Analysis -- A case study. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.1905.05598>
- Sarstedt, M., Ringle, C. M., & Hair, J. F. (2017). *Partial least squares structural equation modeling*. In Springer eBooks (pp. 1–40). https://doi.org/10.1007/978-3-319-05542-8_15-1
- Security Commission Malaysia (2025, Oct 3). *Malaysia launches National Strategy for Financial Literacy 2026-2030 (NS2.0) & Financial Literacy Month (FLM) 2025 for a Resilient Financial Future*. Security Commission Malaysia. <https://www.sc.com.my/resources/media/media-release/malaysia-launches-national-strategy-for-financial-literacy-2026-2030-ns20-financial-literacy-month-flm-2025-for-a-resilient-financial-future>
- Sekhon, H., Ennew, C., Kharouf, H., & Devlin, J. (2014). Trustworthiness and trust: Influences and implications. *Journal of Marketing Management*, 30(3–4), 409–430. <https://doi.org/10.1080/0267257X.2013.842609>
- Shah, P. (2025, Apr 1). How financial services can evolve to stay relevant for Gen Z. EY. https://www.ey.com/en_in/insights/financial-services/how-financial-services-can-evolve-to-stay-relevant-for-gen-z

- Shaikh, I. M., & Amin, H. (2024). Influence of innovation diffusion factors on non-users' adoption of digital banking services in the banking 4.0 era. *Information Discovery and Delivery*. <https://doi.org/10.1108/idd-05-2023-0044>
- Shmueli, G., Sarstedt, M., Hair, J. F., Cheah, J., Ting, H., Vaithilingam, S., & Ringle, C. M. (2019). Predictive model assessment in PLS-SEM: guidelines for using PLSpredict. *European Journal of Marketing*, 53(11), 2322–2347. <https://doi.org/10.1108/ejm-02-2019-0189>
- Simkus, J. (2023, July 31). Pilot Study in Research: Definition & Examples. Simply Psychology. <https://www.simplypsychology.org/pilot-studies.html>
- Singh, P., Roy, N., Bhatt, A. S., Sadual, M. K., & Sahai, A. (2024). Consumer's perspective towards adoption of artificial intelligence: An empirical study among banking companies. *Pacific Business Review (International)*, 17(3), 131-138.
- Solarz, M., & Adamek, J. (2021). Factors affecting mobile banking adoption in Poland: an empirical study. *European Research Studies Journal*, XXIV(Issue 4), 1018–1046. <https://doi.org/10.35808/ersj/2648>
- SPSS Analysis (n.d.). Cronbach's Alpha using SPSS. SPSS Analysis. <https://spssanalysis.com/cronbachs-alpha-in-spss/>
- Susanto, A., Chang, Y., & Ha, Y. (2016). Determinants of continuance intention to use the smartphone banking services. *Industrial Management & Data Systems*, 116(3), 508–525. <https://doi.org/10.1108/imds-05-2015-0195>

- Szajna, B. (1994). Software Evaluation and choice: Predictive validation of the Technology Acceptance instrument. *MIS Quarterly*, 18(3), 319.
<https://doi.org/10.2307/249621>
- Taylor, S., & Todd, P. A. (1995). Understanding information technology usage: A test of competing models. *Information Systems Research*, 6(2), 144–176.
<https://doi.org/10.1287/isre.6.2.144>
- Teo, T. S. H., Srivastava, S. C., & Jiang, L. (2008). Trust and electronic government success: An empirical study. *Journal of Management Information Systems*, 25(3), 99–132. <https://doi.org/10.2753/MIS0742-1222250303>
- Thompson, R. L., Higgins, C. A., & Howell, J. M. (1991). Personal computing: Toward A conceptual model of utilization. *MIS Quarterly*, 15(1), 125–143.
<https://doi.org/10.2307/249443>
- Tjiptono, F., Khan, G., Yeong, E. S., & Kunchambo, V. (2020). Generation Z in Malaysia: The four ‘E’ generation. In *Emerald Publishing Limited eBooks* (pp. 149–163). <https://doi.org/10.1108/978-1-80043-220-820201015>
- Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4), 297–323.
<https://doi.org/10.1007/bf00122574>
- UCLA (n.d.). *What Does Cronbach’s Alpha Mean?* UCLA: Advanced Research Computing – Statistical Methods and Data Analytics.
<https://stats.oarc.ucla.edu/spss/faq/what-does-cronbachs-alpha-mean/>

- Venkatesh, N., Morris, N., Davis, N., & Davis, N. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425. <https://doi.org/10.2307/30036540>
- Vijai, C. (2019). Artificial intelligence in Indian banking sector: Challenges and opportunities. *International Journal of Advanced Research*, 7(4), 1581–1587. <https://doi.org/10.21474/ijar01/8987>
- Watkins, M. W. (2018). Exploratory Factor Analysis: A guide to best practice. *Journal of Black Psychology*, 44(3), 219–246. <https://doi.org/10.1177/0095798418771807>
- Wigley, B. & Kantaria, R. (2023, Nov 17). Equity, diversity and inclusion - Speaking Gen Z: How banks can attract young customers. World Economic Forum. <https://www.weforum.org/stories/2023/11/gen-z-banking-finance-money-trends/>
- Willemsen, M. C., Böckenholt, U., & Johnson, E. J. (2011). Choice by value encoding and value construction: Processes of loss aversion. *Journal of Experimental Psychology General*, 140(3), 303–324. <https://doi.org/10.1037/a0023493>
- Williams, B., Onsman, A., & Brown, T. (2010). Exploratory Factor Analysis: A five-step guide for novices. *Australasian Journal of Paramedicine*, 8, 1–13. <https://doi.org/10.33151/ajp.8.3.93>
- Yahaya, S. N., Bakar, M. H., Bakri, M. H., Jaafar, J. A., & Hishamudin, N. S. (2025). Unveiling the digital revolution: Assessing the dynamic impact of E-banking

adoption among generations Y and Z in Malaysia. *Journal of Ecohumanism*, 4(4), 43–56. <https://doi.org/10.62754/joe.v4i4.6703>

Zahid, A. Z. M. (2024, July 16). *Adnan Zaylani Mohamad Zahid: Banking in the era of generative AI*. <https://www.bis.org/review/r240716g.html>

APPENDICES

Appendix 3.1: Questionnaire question construct

<i>Variable</i>	<i>Item</i>	<i>Description</i>	<i>Sources</i>
Demographic	D1	Gender (Male / Female)	Lockett et al. (1997), Huh et al. (2008) and Hu et al. (2019)
	D2	Age (14-17 / 18-21 / 22-25 / 26-29)	
	D3	Employ Status (Employed / Unemployed / Student)	
	D4	Education (High School / Bachelor degree / Master degree / Doctoral Degree / Diploma or equivalent / Other)	
	D5	AI usage per week (0-5 / 6-10 / 11-15 / 16-20 / >20)	
Perceived Usefulness (PU)	PU1	Using AI in banking can meet my service needs.	Lockett et al. (1997), Huh et al. (2008) and Hu et al. (2019)
	PU2	Banking AI can save time.	
	PU3	Banking AI can improve efficiency.	
	PU4	Overall, I find AI in banking to be useful.	
Perceived Ease of Use (PEOU)	PEOU1	Learning to use AI in banking is easy to me.	Belanche et al. (2019) and Rahaman et al. (2021)
	PEOU2	I find it easy to use AI in banking.	
	PEOU3	I find it easy to become skilled at using banking AI.	
	PEOU4	Interacting with AI in banking requires minimal mental effort from me.	
Attitude (ATT)	ATT1	I believe using AI in banking is a good idea.	Grabner-Kräuter and Faullant (2008), Hu et al. (2019), and Rahman et al. (2021)
	ATT2	Using banking AI is a pleasant experience.	
	ATT3	Utilising AI in banking appears to be a wise idea.	
	ATT4	I am interested in the application of AI in Banking.	
Perceived Risk (PR)	PR1	I believe that money is easy to be stolen by using AI in banking.	Featherman and Pavlou (2003), Grabner-Kräuter and Faullant (2008), Alalwan et al. (2017)
	PR2	I believe personal privacy will be disclosed by using AI in banking.	
	PR3	Using AI in banking subjects my banking account to potential fraud.	
	PR4	AI in banking may not perform well and could create issues with my bank account.	
Perceived Trust (PT)	PT1	I believe that banking AI are trustworthy.	Payne et al. (2018), Susanto et al. (2016), Hassan and Wood (2019), and Lee and Chen (2022)
	PT2	The AI in banking provides banking services in my best interest.	
	PT3	I believe that AI in banking provide safe services.	
	PT4	I trust banking AI to protect my personal information.	
Intention to Adopt AI in Banking	INT1	I would like to use banking AI soon.	Patel et al. (2017), Belanche et al. (2019)
	INT2	If AI in banking become available, I would adopt them.	
	INT3	I intend to use AI banking in my banking process.	

Appendix 3.2: Cronbach's alpha threshold

$\alpha \geq 0.9$	Excellent
$0.9 > \alpha \geq 0.8$	Good
$0.8 > \alpha \geq 0.7$	Acceptable
$0.7 > \alpha \geq 0.6$	Questionable
$0.6 > \alpha \geq 0.5$	Poor
$0.5 > \alpha$	Unacceptable

Cronbach's alpha formula:

$$\alpha = \frac{N\bar{c}}{\bar{v} + (N - 1)\bar{c}}$$

Where: N denotes the number of items, $\bar{c}$ denotes the average inter-item covariance among the items, and $\bar{v}$ denotes the average variance (UCLA, n.d.).

Pilot Test Reliability Analysis

All Variables

Case Processing Summary

		N	%
Cases	Valid	34	100.0
	Excluded ^a	0	.0
	Total	34	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.797	23

Perceived Usefulness

Case Processing Summary

		N	%
Cases	Valid	34	100.0
	Excluded ^a	0	.0
	Total	34	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.811	4

Perceived Ease of Use

Case Processing Summary

		N	%
Cases	Valid	34	100.0
	Excluded ^a	0	.0
	Total	34	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.821	4

Attitude Towards AI in Banking

Case Processing Summary

		N	%
Cases	Valid	34	100.0
	Excluded ^a	0	.0
	Total	34	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.776	4

Perceived Risk

Case Processing Summary

		N	%
Cases	Valid	34	100.0
	Excluded ^a	0	.0
	Total	34	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.840	4

Perceived Trust

Case Processing Summary

		N	%
Cases	Valid	34	100.0
	Excluded ^a	0	.0
	Total	34	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.852	4

Intention to Adopt AI in Banking

Case Processing Summary

		N	%
Cases	Valid	34	100.0
	Excluded ^a	0	.0
	Total	34	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.892	3

Model fit measures summary

	Saturated Model	Estimated Model
SRMR	0.053	0.090
d_ ULS	0.762	2.249

d_G	0.318	0.369
Chi-square	1201.516	1250.051
NFI	0.876	0.871

R-square overview

	R-square	Adjusted R-square
ATT	0.614	0.611
INT	0.599	0.597
PU	0.380	0.379